

## 29. Layer 3 Ethernet Switch Module (ESWM)

### 29.1 Overview

The Layer 3 Ethernet Switch Module (ESWM) consists of an Ethernet switch with higher-level routing capability and multi-protocol interface support. It allows autonomous frame routing within the same and between different network interface protocols (for now only Ethernet) for optimized gateway applications.

Table 29.1 lists the ESWM specifications and Figure 29.1 shows the ESWM block diagram.

**Table 29.1 ESWM specifications**

| Item             | Function                                                  |                                                   | Specification             |
|------------------|-----------------------------------------------------------|---------------------------------------------------|---------------------------|
| Basic function   | PHY interface                                             | Number or port                                    | 2 ports                   |
|                  |                                                           | Type                                              | MII, RMII, GMII, RGMII    |
|                  |                                                           | Speed                                             | 10 Mbps, 100 Mbps, 1 Gbps |
|                  |                                                           | Mode                                              | Full Duplex               |
|                  |                                                           | Management IF(MDIO)                               | Yes                       |
|                  |                                                           | Energy Efficient Ethernet (low power idle)        | Yes                       |
|                  |                                                           | Magic Packet™                                     | Yes                       |
|                  | Switch function                                           | Layer 3 support                                   | Yes                       |
|                  |                                                           | Cut through                                       | Yes                       |
|                  |                                                           | VLAN                                              | Yes                       |
|                  |                                                           | Jumbo frame support                               | Max 64 KB                 |
|                  |                                                           | MAC table                                         | 2048                      |
|                  |                                                           | VLAN table                                        | 4096                      |
|                  |                                                           | Stream (perfect filter, layer 3, layer 2) table   | 256                       |
|                  |                                                           | Layer 2/Layer 3 update rule table                 | 256                       |
|                  | gPTP                                                      | Number of channels                                | 2                         |
|                  | PTP pulse generator                                       | Number of channels                                | 4                         |
| Ethernet AVB     | IEEE802.1BA                                               | The AVB profile                                   | Yes                       |
|                  | IEEE1722-2011                                             | Media transport protocol for audio and video      | Yes                       |
|                  | IEEE 802.1AS-2011                                         | gPTP (a profile of IEEE1588 for AVB systems)      | Yes                       |
|                  | IEEE 802.1Qav-2009                                        | Credit based shaper for bandwidth reservation     | Yes                       |
| Ethernet TSN     | IEEE802.1AS-Rev                                           | Adds redundancy and enhancements to gPTP          | Yes                       |
|                  | IEEE 802.1Qbv-2015                                        | Time aware shaper                                 | Yes                       |
|                  | IEEE 802.1Qbu-2016 (802.3br)                              | Frame preemption                                  | Yes                       |
|                  | IEEE 802.1Qch-2017                                        | Cyclic queuing and forwarding                     | Yes                       |
|                  | IEEE 802.1Qcr                                             | Asynchronous traffic shaping                      | Yes                       |
|                  | IEEE 802.1Qci-2017                                        | Stream based filtering and policing               | Yes (without gate filter) |
|                  | IEEE 802.1CB-2017                                         | Frame replication and elimination for reliability | Yes                       |
| TrustZone Filter | Security attribution and privilege attribution can be set |                                                   |                           |

Note: Please contact a Renesas Electronics sales office for network support.

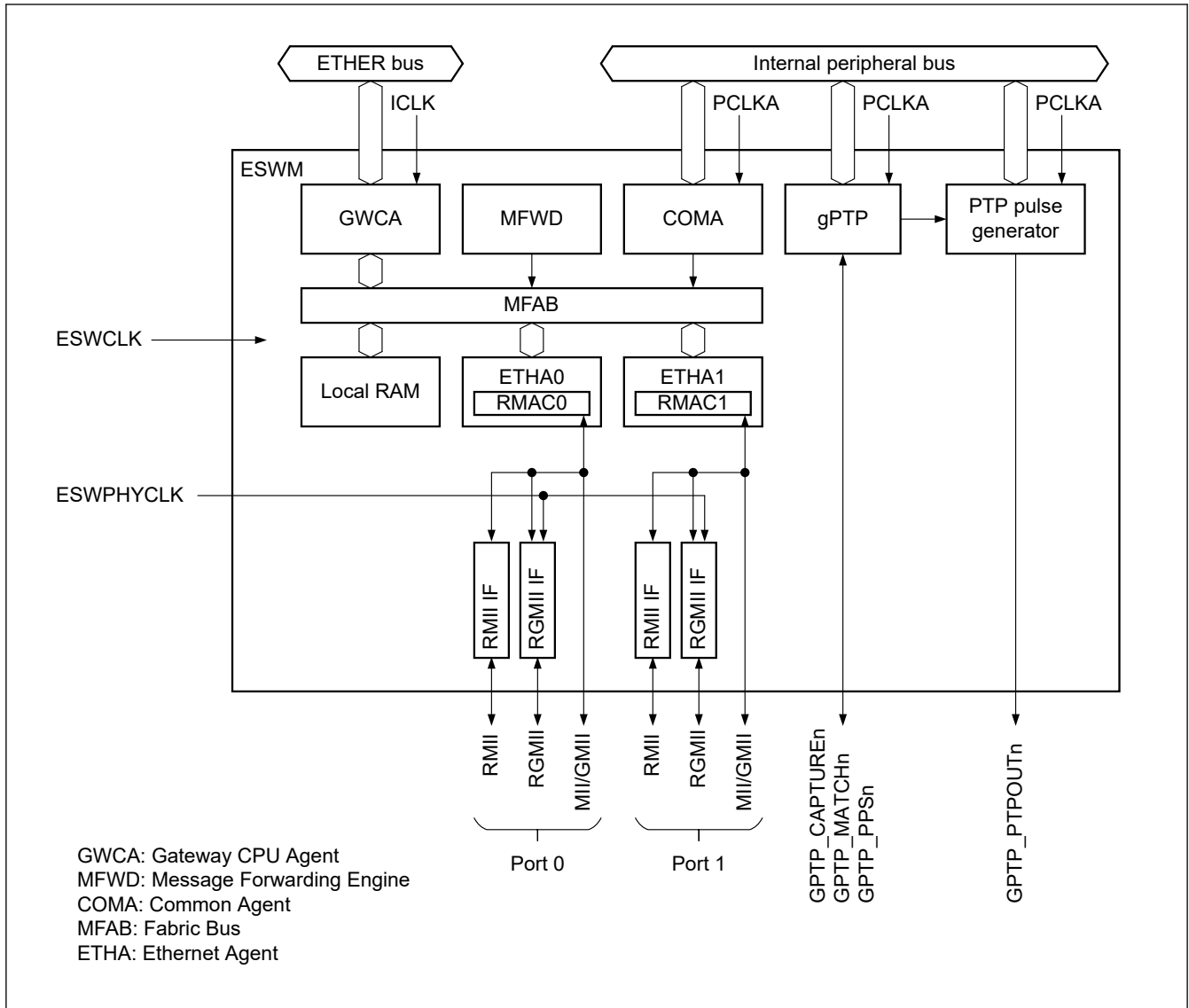


Figure 29.1 ESWM block diagram

Table 29.2 ESWM functional blocks (1 of 2)

| Block name | Function                                                                                                                                                                                                                                                                                                                                                  |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MFWD       | Ethernet Message Forwarding Engine (MFWD) is part of ESWM and aims to filter, forward, and route the switch frames. MFWD snoops the frame information from fabric to redistribute them to the agent after filtering, forwarding, and routing mechanisms.                                                                                                  |
| MFAB       | Fabric Bus (MFAB) is an internal bus used to exchange data between Ethernet Agent [ETHA], Ethernet Common Agent [COMA], Ethernet CPU Agent [GWCA], local RAM, TAG RAM, pointer RAM, and Ethernet Message Forwarding Engine [MFWD].                                                                                                                        |
| ETHA       | Ethernet Agent (ETHA) consists of an agent interface module to allow communication within ESWM. ETHA handles data exchange between ESWM and an Ethernet PHY.                                                                                                                                                                                              |
| RMAC       | Ethernet MAC (RMAC) is an Ethernet controller that conforms to the definition of the MAC (Media Access Control) layer for Ethernet in the IEEE 802.3 standard. By connecting with a physical-layer LSI chip (PHY-LSI) that complies with the standard, RMAC can transmit and receive Ethernet (IEEE 802.3) frames. RMAC has a single MAC layer interface. |
| GWCA       | Ethernet CPU Agent (GWCA) is a CPU interface that consists of an agent interface module allowing communication within ESWM. GWCA handles data exchange between ESWM and the GWCPU subsystem. Software is required for the configuration of Gateway AXI bus interface.                                                                                     |
| COMA       | Ethernet Common Agent (COMA) is an agent of ESWM which aims at gathering common functionalities for other agents. COMA handles the APB to SFR bus conversion, the buffer pointer release for rejected frames and the pointer handling for local RAM. Software is required for the configuration of Ethernet Common Agent.                                 |

**Table 29.2 ESWM functional blocks (2 of 2)**

| Block name | Function                                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| gPTP       | Ethernet Generic PTP timer (GPTP) enables accurate synchronization of the clock in the control system.                               |
| plsotim    | The pulse output timer module (plsotim), a submodule of gPTP, produces a pulsed output according to the base period of a gPTP timer. |

## 29.2 Register Descriptions

### 29.2.1 ESWM Interface Control Registers

#### 29.2.1.1 MIIRR : Media-independent Interface Reset Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x19400

|                    |    |    |    |    |    |    |            |            |    |    |    |    |    |    |            |            |
|--------------------|----|----|----|----|----|----|------------|------------|----|----|----|----|----|----|------------|------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25         | 24         | 23 | 22 | 21 | 20 | 19 | 18 | 17         | 16         |
| Bit field:         | —  | —  | —  | —  | —  | —  | —          | —          | —  | —  | —  | —  | —  | —  | —          | —          |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9          | 8          | 7  | 6  | 5  | 4  | 3  | 2  | 1          | 0          |
| Bit field:         | —  | —  | —  | —  | —  | —  | RMRS<br>T1 | RMRS<br>T0 | —  | —  | —  | —  | —  | —  | RGRS<br>T1 | RGRS<br>T0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          |

| Bit   | Symbol | Function                                               | R/W |
|-------|--------|--------------------------------------------------------|-----|
| 0     | RGRST0 | RGMII0 Reset<br>0: Reset<br>1: Enable                  | R/W |
| 1     | RGRST1 | RGMII1 Reset<br>0: Reset<br>1: Enable                  | R/W |
| 7:2   | —      | These bits are read as 0. The write value should be 0. | R/W |
| 8     | RMRST0 | RMII0 Reset<br>0: Reset<br>1: Enable                   | R/W |
| 9     | RMRST1 | RMII1 Reset<br>0: Reset<br>1: Enable                   | R/W |
| 31:10 | —      | These bits are read as 0. The write value should be 0. | R/W |

### 29.2.1.2 MIICR0 : Media-independent Interface Control Register 0

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x19404

|                    |    |    |    |        |    |    |    |        |    |    |    |    |    |    |             |    |
|--------------------|----|----|----|--------|----|----|----|--------|----|----|----|----|----|----|-------------|----|
| Bit position:      | 31 | 30 | 29 | 28     | 27 | 26 | 25 | 24     | 23 | 22 | 21 | 20 | 19 | 18 | 17          | 16 |
| Bit field:         | —  | —  | —  | —      | —  | —  | —  | —      | —  | —  | —  | —  | —  | —  | —           | —  |
| Value after reset: | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  |
| Bit position:      | 15 | 14 | 13 | 12     | 11 | 10 | 9  | 8      | 7  | 6  | 5  | 4  | 3  | 2  | 1           | 0  |
| Bit field:         | —  | —  | —  | TXCIDE | —  | —  | —  | DIVSTP | —  | —  | —  | —  | —  | —  | MIISEL[1:0] | —  |
| Value after reset: | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  |

| Bit   | Symbol      | Function                                                                                                  | R/W |
|-------|-------------|-----------------------------------------------------------------------------------------------------------|-----|
| 1:0   | MIISEL[1:0] | Media-independent Interface Select<br>0 0: MII/GMII<br>0 1: RGMII<br>1 0: RMII<br>1 1: Setting prohibited | R/W |
| 7:2   | —           | These bits are read as 0. The write value should be 0.                                                    | R/W |
| 8     | DIVSTP      | Clock Divider Stop<br>0: Enable<br>1: Stop                                                                | R/W |
| 11:9  | —           | These bits are read as 0. The write value should be 0.                                                    | R/W |
| 12    | TXCIDE      | RGMII TxC Internal Delay Enable<br>0: TxC internal delay disable<br>1: TxC internal delay enable          | R/W |
| 31:13 | —           | These bits are read as 0. The write value should be 0.                                                    | R/W |

### 29.2.1.3 MIICR1 : Media-independent Interface Control Register 1

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x19408

|                    |    |    |    |        |    |    |    |        |    |    |    |    |    |    |             |    |
|--------------------|----|----|----|--------|----|----|----|--------|----|----|----|----|----|----|-------------|----|
| Bit position:      | 31 | 30 | 29 | 28     | 27 | 26 | 25 | 24     | 23 | 22 | 21 | 20 | 19 | 18 | 17          | 16 |
| Bit field:         | —  | —  | —  | —      | —  | —  | —  | —      | —  | —  | —  | —  | —  | —  | —           | —  |
| Value after reset: | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  |
| Bit position:      | 15 | 14 | 13 | 12     | 11 | 10 | 9  | 8      | 7  | 6  | 5  | 4  | 3  | 2  | 1           | 0  |
| Bit field:         | —  | —  | —  | TXCIDE | —  | —  | —  | DIVSTP | —  | —  | —  | —  | —  | —  | MIISEL[1:0] | —  |
| Value after reset: | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0      | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  |

| Bit | Symbol      | Function                                                                                                  | R/W |
|-----|-------------|-----------------------------------------------------------------------------------------------------------|-----|
| 1:0 | MIISEL[1:0] | Media-independent Interface Select<br>0 0: MII/GMII<br>0 1: RGMII<br>1 0: RMII<br>1 1: Setting prohibited | R/W |
| 7:2 | —           | These bits are read as 0. The write value should be 0.                                                    | R/W |

| Bit   | Symbol | Function                                                                                         | R/W |
|-------|--------|--------------------------------------------------------------------------------------------------|-----|
| 8     | DIVSTP | Clock Divider Stop<br>0: Enable<br>1: Stop                                                       | R/W |
| 11:9  | —      | These bits are read as 0. The write value should be 0.                                           | R/W |
| 12    | TXCIDE | RGMII TXC Internal Delay Enable<br>0: TXC Internal delay disable<br>1: TXC Internal delay enable | R/W |
| 31:13 | —      | These bits are read as 0. The write value should be 0.                                           | R/W |

#### 29.2.1.4 MCCESR : Media Clock Capture Event Select Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x19410

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |            |            |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|------------|------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17         | 16         |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —          | —          |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1          | 0          |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | MCCE<br>S1 | MCCE<br>S0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0          | 0          |

| Bit  | Symbol | Function                                                    | R/W |
|------|--------|-------------------------------------------------------------|-----|
| 0    | MCCES0 | Media Clock 0 Capture Event Select<br>0: I/O Port<br>1: ELC | R/W |
| 1    | MCCES1 | Media Clock 1 Capture Event Select<br>0: I/O Port<br>1: ELC | R/W |
| 31:2 | —      | These bits are read as 0. The write value should be 0.      | R/W |

#### 29.2.1.5 TASSTSR : TAS Status Monitor Signal Select Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x19420

|                    |    |    |    |           |    |    |    |    |    |    |           |    |    |    |    |    |
|--------------------|----|----|----|-----------|----|----|----|----|----|----|-----------|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28        | 27 | 26 | 25 | 24 | 23 | 22 | 21        | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | MSS3[4:0] |    |    |    | —  | —  | —  | MSS2[4:0] |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12        | 11 | 10 | 9  | 8  | 7  | 6  | 5         | 4  | 3  | 2  | 1  | 0  |
| Bit field:         | —  | —  | —  | MSS1[4:0] |    |    |    | —  | —  | —  | MSS0[4:0] |    |    |    |    |    |
| Value after reset: | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  | 0  | 0         | 0  | 0  | 0  | 0  | 0  |

| Bit | Symbol    | Function                                                                                                                                            | R/W |
|-----|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 4:0 | MSS0[4:0] | ET_TAS_STA0 Monitor Signal Select<br>Select signal to output ET_TAS_STA0 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0] | R/W |
| 7:5 | —         | These bits are read as 0. The write value should be 0.                                                                                              | R/W |

| Bit   | Symbol    | Function                                                                                                                                            | R/W |
|-------|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 12:8  | MSS1[4:0] | ET_TAS_STA1 Monitor Signal Select<br>Select signal to output ET_TAS_STA1 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0] | R/W |
| 15:13 | —         | These bits are read as 0. The write value should be 0.                                                                                              | R/W |
| 20:16 | MSS2[4:0] | ET_TAS_STA2 Monitor Signal Select<br>Select signal to output ET_TAS_STA2 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0] | R/W |
| 23:21 | —         | These bits are read as 0. The write value should be 0.                                                                                              | R/W |
| 28:24 | MSS3[4:0] | ET_TAS_STA3 Monitor Signal Select<br>Select signal to output ET_TAS_STA3 pin from race_etha0_tas_gate_state[8:0] and race_etha1_tas_gate_state[8:0] | R/W |
| 31:29 | —         | These bits are read as 0. The write value should be 0.                                                                                              | R/W |

## 29.2.2 Register Descriptions of Related ESWM Interrupts

### 29.2.2.1 TPEMIMC1 : Error and Monitoring Interrupt Mapping Configuration Register 1

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0004

|                    |    |            |    |    |    |    |    |           |      |    |            |    |    |    |    |           |      |
|--------------------|----|------------|----|----|----|----|----|-----------|------|----|------------|----|----|----|----|-----------|------|
| Bit position:      | 31 | 30         | 29 | 28 | 27 | 26 | 25 | 24        | 23   | 22 | 21         | 20 | 19 | 18 | 17 | 16        |      |
| Bit field:         | —  | CSICM[2:0] |    |    |    | —  | —  | CSIG<br>M | CSIM | —  | CEICM[2:0] |    |    | —  | —  | CEIG<br>M | CEIM |
| Value after reset: | 0  | 0          | 0  | 0  | 0  | 0  | 0  | 0         | 0    | 0  | 0          | 0  | 0  | 0  | 0  | 0         |      |
| Bit position:      | 15 | 14         | 13 | 12 | 11 | 10 | 9  | 8         | 7    | 6  | 5          | 4  | 3  | 2  | 1  | 0         |      |
| Bit field:         | —  | FSICM[2:0] |    |    |    | —  | —  | FSIGM     | FSIM | —  | FEICM[2:0] |    |    | —  | —  | FEIGM     | FEIM |
| Value after reset: | 0  | 0          | 0  | 0  | 0  | 0  | 0  | 0         | 0    | 0  | 0          | 0  | 0  | 0  | 0  | 0         |      |

| Bit | Symbol     | Function                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W |
|-----|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | FEIM       | MFWD Error Interrupt Mapping<br>MFWD Error Interrupt contains the following interrupts: <ul style="list-style-type: none"> <li>• FWEIS0</li> <li>• FWEIS1</li> <li>• FWEIS2</li> <li>• FWEIS3</li> <li>• FWEIS4</li> <li>• FWEIS5</li> <li>• FWEIS6</li> <li>• FWEIS7</li> <li>• FWEIS8</li> </ul> 0: MFWD Error Interrupt is mapped to ETHER_FWEI<br>1: MFWD Error Interrupt is mapped to a specific core depending on TPEMIMC1.FEIGM register | R/W |
| 1   | FEIGM      | MFWD Error Interrupt GWCA Mapping<br>0: When TPEMIMC1.FEIM is set to 1, MFWD Error Interrupt is mapped to ETHER_GWDI[TPEMIMC1.FEICM]<br>1: Setting prohibited                                                                                                                                                                                                                                                                                   | R/W |
| 3:2 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                          | R/W |
| 6:4 | FEICM[2:0] | MFWD Error Interrupt Core Mapping<br>Determine to which pin MFWD Error Interrupt is mapped when mapped to a GWCA                                                                                                                                                                                                                                                                                                                                | R/W |
| 7   | —          | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                             | R/W |

| Bit   | Symbol     | Function                                                                                                                                                                                                                                                                                                                           | R/W |
|-------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 8     | FSIM       | MFWD Status Interrupt Mapping<br>MFWD Status Interrupt contains the following interrupt:<br><ul style="list-style-type: none"> <li>FWMIS0 [MFWD]</li> </ul> 0: MFWD Status Interrupt is mapped to ETHER_FWSI<br>1: MFWD Status Interrupt is mapped to a specific core depending on TPEMIMC1.FSIGM register                         | R/W |
| 9     | FSIGM      | MFWD Status Interrupt GWCA Mapping<br>0: When TPEMIMC1.FSIM is set to 1, MFWD Status Interrupt is mapped to ETHER_GWDI[TPEMIMC1.FSICM]<br>1: Setting prohibited                                                                                                                                                                    | R/W |
| 11:10 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                             | R/W |
| 14:12 | FSICM[2:0] | MFWD Status Interrupt Core Mapping<br>Determine to which pin MFWD Status Interrupt is mapped when mapped to a GWCA                                                                                                                                                                                                                 | R/W |
| 15    | —          | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                | R/W |
| 16    | CEIM       | COMA Error Interrupt Mapping<br>COMA Error Interrupt contains the following interrupts:<br><ul style="list-style-type: none"> <li>CAEIS0 [COMA]</li> <li>CAEIS1 [COMA]</li> </ul> 0: COMA Error Interrupt is mapped to ETHER_CAEI<br>1: COMA Error Interrupt is mapped to a specific core depending on TPEMIMC1.CEIGM register     | R/W |
| 17    | CEIGM      | COMA Error Interrupt GWCA Mapping<br>0: When TPEMIMC1.CEIM is set to 1, COMA Error Interrupt is mapped to ETHER_GWDI[TPEMIMC1.CEICM]<br>1: Setting prohibited                                                                                                                                                                      | R/W |
| 19:18 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                             | R/W |
| 22:20 | CEICM[2:0] | COMA Error Interrupt Core Mapping<br>Determine to which pin COMA Error Interrupt is mapped when mapped to a GWCA                                                                                                                                                                                                                   | R/W |
| 23    | —          | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                | R/W |
| 24    | CSIM       | COMA Status Interrupt Mapping<br>COMA Status Interrupt contains the following interrupts:<br><ul style="list-style-type: none"> <li>CAMIS0 [COMA]</li> <li>CAMIS1 [COMA]</li> </ul> 0: COMA Status Interrupt is mapped to ETHER_CAMI<br>1: COMA Status Interrupt is mapped to a specific core depending on TPEMIMC1.CSIGM register | R/W |
| 25    | CSIGM      | COMA Status Interrupt GWCA Mapping<br>0: When TPEMIMC1.CSIM is set to 1, COMA Status Interrupt is mapped to ETHER_GWDI[TPEMIMC1.CSICM]<br>1: Setting prohibited                                                                                                                                                                    | R/W |
| 27:26 | —          | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                             | R/W |
| 30:28 | CSICM[2:0] | COMA Status Interrupt Core Mapping<br>Used to decide to which pin COMA Status Interrupt are mapped when mapped to a GWCA.                                                                                                                                                                                                          | R/W |
| 31    | —          | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                | R/W |

### 29.2.2.2 TPIMIMC2 : Error and Monitoring Interrupt Mapping Configuration Register 2

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0008

|                    |    |    |    |    |    |    |    |    |    |             |    |    |    |    |        |       |
|--------------------|----|----|----|----|----|----|----|----|----|-------------|----|----|----|----|--------|-------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22          | 21 | 20 | 19 | 18 | 17     | 16    |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —           | —  | —  | —  | —  | —      | —     |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6           | 5  | 4  | 3  | 2  | 1      | 0     |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | GEICM0[2:0] |    |    | —  | —  | GEIGM0 | GEIM0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |

| Bit  | Symbol      | Function                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W |
|------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | GEIM0       | GWCA0 Error Interrupt Mapping<br>GWCA0 Error Interrupt contains the following interrupts: <ul style="list-style-type: none"> <li>GWEIS0 except GWEIS0.TDFES for GWCA0 [GWCA]</li> <li>GWEIS1 for GWCA0 [GWCA]</li> <li>GWEIS4 for GWCA0 [GWCA]</li> <li>GWEIS5 for GWCA0 [GWCA]</li> </ul> 0: GWCA0 Error Interrupt is mapped to ETHER_GWEI<br>1: GWCA0 Error Interrupt is mapped to a specific core depending on TPIMIMC2.GEIGM0 register | R/W |
| 1    | GEIGM0      | GWCA0 Error Interrupt GWCA Mapping<br>0: When TPIMIMC2.GEIM0 is set to 1, GWCA0 Error Interrupt is mapped to ETHER_GWDI[TPIMIMC2.GEICM0]<br>1: Setting prohibited                                                                                                                                                                                                                                                                          | R/W |
| 3:2  | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                     | R/W |
| 6:4  | GEICM0[2:0] | GWCA0 Error Interrupt Core Mapping<br>Determine to which pin GWCA0 Error Interrupt is mapped when mapped to a GWCA.                                                                                                                                                                                                                                                                                                                        | R/W |
| 31:7 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                                     | R/W |

### 29.2.2.3 TPIMIMC3 : Error and Monitoring Interrupt Mapping Configuration Register 3

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x000C

|                    |    |             |    |    |    |    |        |       |    |             |    |    |    |    |        |       |
|--------------------|----|-------------|----|----|----|----|--------|-------|----|-------------|----|----|----|----|--------|-------|
| Bit position:      | 31 | 30          | 29 | 28 | 27 | 26 | 25     | 24    | 23 | 22          | 21 | 20 | 19 | 18 | 17     | 16    |
| Bit field:         | —  | —           | —  | —  | —  | —  | —      | —     | —  | —           | —  | —  | —  | —  | —      | —     |
| Value after reset: | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |
| Bit position:      | 15 | 14          | 13 | 12 | 11 | 10 | 9      | 8     | 7  | 6           | 5  | 4  | 3  | 2  | 1      | 0     |
| Bit field:         | —  | ESICM0[2:0] |    |    | —  | —  | ESIGM0 | ESIM0 | —  | EEICM0[2:0] |    |    | —  | —  | EEIGM0 | EEIM0 |
| Value after reset: | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |



| Bit   | Symbol      | Function                                                                                                                                                                                                                                                                                                                                                                                                              | R/W |
|-------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | EEIM0       | ETHA0 Error Interrupt Mapping<br>ETHA0 Error Interrupt contains the following interrupts: <ul style="list-style-type: none"> <li>EAEIS0 for ETHA0 [ETHA]</li> <li>EAEIS1 for ETHA0 [ETHA]</li> <li>EAEIS2 for ETHA0 [ETHA]</li> <li>MEIS for RMAC0 [RMAC]</li> </ul> 0: ETHA0 Error Interrupt is mapped to ETHER_EAEI0<br>1: ETHA0 Error Interrupt is mapped to a specific core depending on TPEMIMC3.EEIGM0 register | R/W |
| 1     | EEIGM0      | ETHA0 Error Interrupt GWCA Mapping<br>0: When TPEMIMC3.EEIM0 is set to 1, ETHA0 Error Interrupt is mapped to ETHER_GWDI[TPEMIMC3.EEICM0]<br>1: Setting prohibited                                                                                                                                                                                                                                                     | R/W |
| 3:2   | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                | R/W |
| 6:4   | EEICM0[2:0] | ETHA0 Error Interrupt Core Mapping<br>Determine to which pin ETHA0 Error Interrupt is mapped when mapped to a GWCA.                                                                                                                                                                                                                                                                                                   | R/W |
| 7     | —           | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                   | R/W |
| 8     | ESIM0       | ETHA0 Status Interrupt Mapping<br>ETHA0 Status Interrupt contains the following interrupt: <ul style="list-style-type: none"> <li>MMIS0 for RMAC 0 [RMAC]</li> </ul> 0: ETHA0 Status Interrupt is mapped to ETHER_EASI0<br>1: ETHA0 Status Interrupt is mapped to a specific core depending on TPEMIMC3.ESIGM0 register                                                                                               | R/W |
| 9     | ESIGM0      | ETHA0 Status Interrupt GWCA Mapping<br>0: When TPEMIMC3.ESIM0 is set to 1, ETHA0 Status Interrupt is mapped to ETHER_GWDI[TPEMIMC3.ESICM0]<br>1: Setting prohibited                                                                                                                                                                                                                                                   | R/W |
| 11:10 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                | R/W |
| 14:12 | ESICM0[2:0] | ETHA0 Status Interrupt Core Mapping<br>Determine to which pin ETHA0 Status Interrupt is mapped when mapped to a GWCA.                                                                                                                                                                                                                                                                                                 | R/W |
| 31:15 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                                                                                                | R/W |

#### 29.2.2.4 TPEMIMC4 : Error and Monitoring Interrupt Mapping Configuration Register 4

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0010

|                    |    |             |    |    |    |    |        |       |    |             |    |    |    |    |        |       |
|--------------------|----|-------------|----|----|----|----|--------|-------|----|-------------|----|----|----|----|--------|-------|
| Bit position:      | 31 | 30          | 29 | 28 | 27 | 26 | 25     | 24    | 23 | 22          | 21 | 20 | 19 | 18 | 17     | 16    |
| Bit field:         | —  | —           | —  | —  | —  | —  | —      | —     | —  | —           | —  | —  | —  | —  | —      | —     |
| Value after reset: | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |
| Bit position:      | 15 | 14          | 13 | 12 | 11 | 10 | 9      | 8     | 7  | 6           | 5  | 4  | 3  | 2  | 1      | 0     |
| Bit field:         | —  | ESICM1[2:0] |    |    | —  | —  | ESIGM1 | ESIM1 | —  | EEICM1[2:0] |    |    | —  | —  | EEIGM1 | EEIM1 |
| Value after reset: | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     | 0  | 0           | 0  | 0  | 0  | 0  | 0      | 0     |

| Bit | Symbol | Function                                                                                                                                                                                                                                                                                                                                                                                                             | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | EEIM1  | ETHA1 Error Interrupt Mapping<br>ETHA1 Error Interrupt contains the following interrupt: <ul style="list-style-type: none"> <li>EAEIS0 for ETHA1 [ETHA]</li> <li>EAEIS1 for ETHA1 [ETHA]</li> <li>EAEIS2 for ETHA1 [ETHA]</li> <li>MEIS for RMAC1 [RMAC]</li> </ul> 0: ETHA1 Error Interrupt is mapped to ETHER_EAEI1<br>1: ETHA1 Error Interrupt is mapped to a specific core depending on TPEMIMC4.EEIGM1 register | R/W |

| Bit   | Symbol      | Function                                                                                                                                                                                                                                                                     | R/W |
|-------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 1     | EEIGM1      | ETHA1 Error Interrupt GWCA Mapping<br>0: When TPEMIMC4.EEIM1 is set to 1, ETHA1 Error Interrupt is mapped to ETHER_GWDI[TPEMIMC4.EEICM1]<br>1: Setting prohibited                                                                                                            | R/W |
| 3:2   | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                       | R/W |
| 6:4   | EEICM1[2:0] | ETHA1 Error Interrupt Core Mapping<br>Determine to which pin ETHA1 Error Interrupt is mapped when mapped to a GWCA.                                                                                                                                                          | R/W |
| 7     | —           | This bit is read as 0. The write value should be 0.                                                                                                                                                                                                                          | R/W |
| 8     | ESIM1       | ETHA1 Status Interrupt Mapping<br>ETHA1 Status Interrupt contains the following interrupt:<br>• MMIS0 for RMAC 1 [RMAC]<br>0: ETHA1 Status Interrupt is mapped to ETHER_EAS1<br>1: ETHA1 Status Interrupt is mapped to a specific core depending on TPEMIMC4.ESIGM1 register | R/W |
| 9     | ESIGM1      | ETHA1 Status Interrupt GWCA Mapping<br>0: When TPEMIMC4.ESIM1 is set to 1, ETHA1 Status Interrupt is mapped to ETHER_GWDI[TPEMIMC4.ESICM1]<br>1: Setting prohibited                                                                                                          | R/W |
| 11:10 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                       | R/W |
| 14:12 | ESICM1[2:0] | ETHA1 Status Interrupt Core Mapping<br>Determine to which pin ETHA1 Status Interrupt is mapped when mapped to a GWCA.                                                                                                                                                        | R/W |
| 31:15 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                       | R/W |

### 29.2.2.5 TPEMIMC6t : Error and Monitoring Interrupt Mapping Configuration Register 6t (t = 0 to 1)

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0080 + 0x04 × t

|                    |    |    |    |    |    |    |    |    |    |    |    |    |              |    |        |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|--------------|----|--------|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19           | 18 | 17     | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —            | —  | —      | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0  | 0      | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3            | 2  | 1      | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | GTSICM0[2:0] |    | GTSIM0 |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0  | 0      | 0  |

| Bit  | Symbol       | Function                                                                                                                                                                                                                                                                                                                               | R/W |
|------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | GTSIM0       | GWCA0 Timestamp Interrupt Mapping<br>GWCA0 Timestamp Interrupt contains the following interrupts:<br>• GWTDIS.TSDIS for GWCA0 [GWCA]<br>• GWEIS0.TDFES for GWCA0 [GWCA]<br>0: GWCA0 Timestamp Interrupt t is mapped to ETHER_TSDI[t]<br>1: GWCA0 Timestamp Interrupt t is mapped to ETHER_GWDIn depending on TPEMIMC6t.GTSICM0 setting | R/W |
| 3:1  | GTSICM0[2:0] | GWCA0 Timestamp Interrupt Core Mapping<br>Map GWCA0 Timestamp Interrupt t to ETHER_GWDI[TPEMIMC6t.GTSICM0] when TPEMIMC6t.GTSIM0 is set.                                                                                                                                                                                               | R/W |
| 31:4 | —            | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                 | R/W |

### 29.2.2.6 TPEMIMC7t : Error and Monitoring Interrupt Mapping Configuration Register 7t (t = 0 to 63)

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0100 + 0x04 × t

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |             |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|-------------|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18          | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —           | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4  | 3  | 2           | 1  | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | GDICM0[2:0] |    |    |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0           | 0  | 0  |

| Bit  | Symbol      | Function                                                                                                                                                                                                                                                                                                                                         | R/W |
|------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2:0  | GDICM0[2:0] | GWCA0 Data Interrupt Core Mapping<br>Map GWCA0 Data Interrupts t to ETHER_GWDI[TPEMIMC7t.GDICM0].<br>RMAC2 Status Interrupt contains the following interrupts: <ul style="list-style-type: none"> <li>• GWDISI.DIS for GWCA0 [GWCA]</li> <li>• GWEIS2i.DFES for GWCA0 [GWCA]</li> <li>• GWEIS3.IAOEst for t = 0 to 4 for GWCA0 [GWCA]</li> </ul> | R/W |
| 31:3 | —           | These bits are read as 0. The write value should be 0.                                                                                                                                                                                                                                                                                           | R/W |

### 29.2.2.7 TSIM : Summarized Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0700

|                    |    |    |    |    |    |    |    |    |    |    |      |      |    |      |     |     |
|--------------------|----|----|----|----|----|----|----|----|----|----|------|------|----|------|-----|-----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21   | 20   | 19 | 18   | 17  | 16  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —    | —    | —  | —    | —   | —   |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0    | 0    | 0  | 0    | 0   | 0   |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5    | 4    | 3  | 2    | 1   | 0   |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | EIM1 | EIM0 | —  | GIM0 | CIM | FIM |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0    | 0    | 0  | 0    | 0   | 0   |

| Bit | Symbol | Function                                                                                                 | R/W |
|-----|--------|----------------------------------------------------------------------------------------------------------|-----|
| 0   | FIM    | MFWD Interrupt Mirroring<br>0: No interrupt is set in MFWD<br>1: An interrupt is set in MFWD             | R   |
| 1   | CIM    | COMA Interrupt Mirroring<br>0: No interrupt is set in COMA agent<br>1: An interrupt is set in COMA agent | R   |
| 2   | GIM0   | GWCA0 Interrupt Monitoring<br>0: No interrupt is set in GWCA0<br>1: An interrupt is set in GWCA0         | R   |
| 3   | —      | This bit is read as 0.                                                                                   | R   |
| 4   | EIM0   | ETHA0 Interrupt Monitoring<br>0: No interrupt is set in ETHA0<br>1: An interrupt is set in ETHA0         | R   |

| Bit  | Symbol | Function                                                                                         | R/W |
|------|--------|--------------------------------------------------------------------------------------------------|-----|
| 5    | EIM1   | ETHA1 Interrupt Monitoring<br>0: No interrupt is set in ETHA1<br>1: An interrupt is set in ETHA1 | R   |
| 31:6 | —      | These bits are read as 0.                                                                        | R   |

### 29.2.2.8 TFIM : MFWD Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0704

|                    |    |    |    |    |    |    |          |          |          |          |          |          |          |          |          |          |
|--------------------|----|----|----|----|----|----|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25       | 24       | 23       | 22       | 21       | 20       | 19       | 18       | 17       | 16       |
| Bit field:         | —  | —  | —  | —  | —  | —  | —        | —        | —        | —        | —        | —        | —        | —        | —        | —        |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9        | 8        | 7        | 6        | 5        | 4        | 3        | 2        | 1        | 0        |
| Bit field:         | —  | —  | —  | —  | —  | —  | FWMISIM0 | FWEISIM8 | FWEISIM7 | FWEISIM6 | FWEISIM5 | FWEISIM4 | FWEISIM3 | FWEISIM2 | FWEISIM1 | FWEISIM0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        | 0        |

| Bit   | Symbol   | Function                                                                                                                                                 | R/W |
|-------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0     | FWEISIM0 | FWEIS0 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS0 register<br>1: An interrupt is set in forwarding engine FWEIS0 register | R   |
| 1     | FWEISIM1 | FWEIS1 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS1 register<br>1: An interrupt is set in forwarding engine FWEIS1 register | R   |
| 2     | FWEISIM2 | FWEIS2 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS2 register<br>1: An interrupt is set in forwarding engine FWEIS2 register | R   |
| 3     | FWEISIM3 | FWEIS3 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS3 register<br>1: An interrupt is set in forwarding engine FWEIS3 register | R   |
| 4     | FWEISIM4 | FWEIS4 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS4 register<br>1: An interrupt is set in forwarding engine FWEIS4 register | R   |
| 5     | FWEISIM5 | FWEIS5 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS5 register<br>1: An interrupt is set in forwarding engine FWEIS5 register | R   |
| 6     | FWEISIM6 | FWEIS6 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS6 register<br>1: An interrupt is set in forwarding engine FWEIS6 register | R   |
| 7     | FWEISIM7 | FWEIS7 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS7 register<br>1: An interrupt is set in forwarding engine FWEIS7 register | R   |
| 8     | FWEISIM8 | FWEIS8 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWEIS8 register<br>1: An interrupt is set in forwarding engine FWEIS8 register | R   |
| 9     | FWMISIM0 | FWMIS0 Interrupt Mirroring<br>0: No interrupt is set in forwarding engine FWMIS0 register<br>1: An interrupt is set in forwarding engine FWMIS0 register | R   |
| 31:10 | —        | These bits are read as 0.                                                                                                                                | R   |

## 29.2.2.9 TCIM : COMA Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0708

|                    |    |    |    |    |    |    |    |    |    |    |    |              |              |              |              |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|--------------|--------------|--------------|--------------|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20           | 19           | 18           | 17           | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —            | —            | —            | —            | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0            | 0            | 0            | 0  |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7  | 6  | 5  | 4            | 3            | 2            | 1            | 0  |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | CAMIS<br>IM1 | CAMIS<br>IM0 | CAEIS<br>IM1 | CAEIS<br>IM0 | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0            | 0            | 0            | 0  |

| Bit  | Symbol   | Function                                                                                                                                   | R/W |
|------|----------|--------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | —        | This bit is read as 0.                                                                                                                     | R   |
| 1    | CAEISIM0 | CAEIS0 Interrupt Mirroring<br>0: No interrupt is set in COMA agent CAEIS0 register<br>1: An interrupt is set in COMA agent CAEIS0 register | R   |
| 2    | CAEISIM1 | CAEIS1 Interrupt Mirroring<br>0: No interrupt is set in COMA agent CAEIS1 register<br>1: An interrupt is set in COMA agent CAEIS1 register | R   |
| 3    | CAMISIM0 | CAMIS0 Interrupt Mirroring<br>0: No interrupt is set in COMA agent CAMIS0 register<br>1: An interrupt is set in COMA agent CAMIS0 register | R   |
| 4    | CAMISIM1 | CAMIS1 Interrupt Mirroring<br>0: No interrupt is set in COMA agent CAMIS1 register<br>1: An interrupt is set in COMA agent CAMIS1 register | R   |
| 31:5 | —        | These bits are read as 0.                                                                                                                  | R   |

## 29.2.2.10 TGIM0 : GWCA0 Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0710

|                    |    |    |    |    |    |    |    |    |              |              |              |              |              |              |               |             |
|--------------------|----|----|----|----|----|----|----|----|--------------|--------------|--------------|--------------|--------------|--------------|---------------|-------------|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23           | 22           | 21           | 20           | 19           | 18           | 17            | 16          |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —            | —            | —            | —            | —            | —            | —             | —           |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0            | 0            | 0            | 0            | 0            | 0             | 0           |
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9  | 8  | 7            | 6            | 5            | 4            | 3            | 2            | 1             | 0           |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | GWEI<br>SIM5 | GWEI<br>SIM4 | GWEI<br>SIM3 | GWEI<br>SIM2 | GWEI<br>SIM1 | GWEI<br>SIM0 | GWTS<br>DISIM | GWDI<br>SIM |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0            | 0            | 0            | 0            | 0            | 0            | 0             | 0           |

| Bit | Symbol     | Function                                                                                                                                     | R/W |
|-----|------------|----------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0   | GWDISIM    | GWDIS Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWDIS register<br>1: An interrupt is set in GWCA0 GWDIS register                | R   |
| 1   | GWTS DISIM | GWTS DISIM Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWTS DISIM register<br>1: An interrupt is set in GWCA0 GWTS DISIM register | R   |

| Bit  | Symbol   | Function                                                                                                                           | R/W |
|------|----------|------------------------------------------------------------------------------------------------------------------------------------|-----|
| 2    | GWEISIM0 | GWEIS0 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS0 register<br>1: An interrupt is set in GWCA0 GWEIS0 register   | R   |
| 3    | GWEISIM1 | GWEIS1 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS1 register<br>1: An interrupt is set in GWCA0 GWEIS1 register   | R   |
| 4    | GWEISIM2 | GWEIS2 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS2 register<br>1: An interrupt is set in GWCA0 GWEIS2 register   | R   |
| 5    | GWEISIM3 | GWEIS3 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS3 register<br>1: An interrupt is set in GWCA0 GWEIS3 register   | R   |
| 6    | GWEISIM4 | GWEIS4 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS4 register<br>1: An interrupt is set in GWCA0 GWEIS4 register   | R   |
| 7    | GWEISIM5 | GWEIS5 Interrupt Mirroring<br>0: No interrupt is set in GWCA0 GWEIS5 register.<br>1: An interrupt is set in GWCA0 GWEIS5 register. | R   |
| 31:8 | —        | These bits are read as 0.                                                                                                          | R   |

### 29.2.2.11 TEIM0 : ETHA0 Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0720

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

|                    |    |    |    |    |    |    |   |   |   |   |   |            |            |              |              |              |
|--------------------|----|----|----|----|----|----|---|---|---|---|---|------------|------------|--------------|--------------|--------------|
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4          | 3          | 2            | 1            | 0            |
| Bit field:         | —  | —  | —  | —  | —  | —  | — | — | — | — | — | MMISI<br>M | MEISI<br>M | EAEIS<br>IM2 | EAEIS<br>IM1 | EAEIS<br>IM0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0          | 0          | 0            | 0            | 0            |

| Bit  | Symbol   | Function                                                                                                                                | R/W |
|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | EAEISIM0 | EAEIS0 Interrupt Mirroring<br>0: No interrupt is set in ETHA0 EAEIS0 register<br>1: An interrupt is set in ETHA0 EAEIS0 register        | R   |
| 1    | EAEISIM1 | EAEIS1 Interrupt Mirroring<br>0: No interrupt is set in ETHA0 EAEIS1 register<br>1: An interrupt is set in ETHA0 EAEIS1 register        | R   |
| 2    | EAEISIM2 | EAEIS2 Interrupt Mirroring<br>0: No interrupt is set in ETHA0 EAEIS2 register<br>1: An interrupt is set in ETHA0 EAEIS2 register        | R   |
| 3    | MEISIM   | MEIS Interrupt Mirroring<br>0: No interrupt is set in ETHA0 RMAC MEIS register<br>1: An interrupt is set in ETHA0 RMAC MEIS register    | R   |
| 4    | MMISIM   | MMIS0 Interrupt Mirroring<br>0: No interrupt is set in ETHA0 RMAC MMIS0 register<br>1: An interrupt is set in ETHA0 RMAC MMIS0 register | R   |
| 31:5 | —        | These bits are read as 0.                                                                                                               | R   |

### 29.2.2.12 TEIM1 : ETHA1 Interrupt Mirroring Register

Base address: ESWM = 0x403C\_8000  
ESWM\_NS = 0x503C\_8000

Offset address: 0x0724

|                    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Bit position:      | 31 | 30 | 29 | 28 | 27 | 26 | 25 | 24 | 23 | 22 | 21 | 20 | 19 | 18 | 17 | 16 |
| Bit field:         | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  | —  |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  |

|                    |    |    |    |    |    |    |   |   |   |   |   |            |            |              |              |              |
|--------------------|----|----|----|----|----|----|---|---|---|---|---|------------|------------|--------------|--------------|--------------|
| Bit position:      | 15 | 14 | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4          | 3          | 2            | 1            | 0            |
| Bit field:         | —  | —  | —  | —  | —  | —  | — | — | — | — | — | MMISI<br>M | MEISI<br>M | EAEIS<br>IM2 | EAEIS<br>IM1 | EAEIS<br>IM0 |
| Value after reset: | 0  | 0  | 0  | 0  | 0  | 0  | 0 | 0 | 0 | 0 | 0 | 0          | 0          | 0            | 0            | 0            |

| Bit  | Symbol   | Function                                                                                                                                | R/W |
|------|----------|-----------------------------------------------------------------------------------------------------------------------------------------|-----|
| 0    | EAEISIM0 | EAEIS0 Interrupt Mirroring<br>0: No interrupt is set in ETHA1 EAEIS0 register<br>1: An interrupt is set in ETHA1 EAEIS0 register        | R   |
| 1    | EAEISIM1 | EAEIS1 Interrupt Mirroring<br>0: No interrupt is set in ETHA1 EAEIS1 register<br>1: An interrupt is set in ETHA1 EAEIS1 register        | R   |
| 2    | EAEISIM2 | EAEIS2 Interrupt Mirroring<br>0: No interrupt is set in ETHA1 EAEIS2 register<br>1: An interrupt is set in ETHA1 EAEIS2 register        | R   |
| 3    | MEISIM   | MEIS Interrupt Mirroring<br>0: No interrupt is set in ETHA1 RMAC MEIS register<br>1: An interrupt is set in ETHA1 RMAC MEIS register    | R   |
| 4    | MMISIM   | MMIS0 Interrupt Mirroring<br>0: No interrupt is set in ETHA1 RMAC MMIS0 register<br>1: An interrupt is set in ETHA1 RMAC MMIS0 register | R   |
| 31:5 | —        | These bits are read as 0.                                                                                                               | R   |

### 29.2.3 ESWM Core Registers

See the following sections:

- [section 30.3. MFWD Register Descriptions](#)
- [section 31.3. COMA Register Descriptions](#)
- [section 32.3. ETHA Register Descriptions](#)
- [section 33.4. RMAC Register Descriptions](#)
- [section 34.3. GWCA Register Descriptions](#)

### 29.2.4 gPTP and PTP Pulse Generator Registers

See the following sections:

- [section 35.3. gPTP Register Descriptions](#)
- [section 35.6.3. Plsotim Register Descriptions](#)

## 29.3 Register Utilization

### 29.3.1 Software Flows

Restrictions on software flows are as follows:

- SW: ESWM registers can only be written using the flows specified in this section

- SW: All the software flows described in this section can only be used sequentially.

### 29.3.1.1 Software Flow Legend

Figure 29.2 shows the software flow legend.

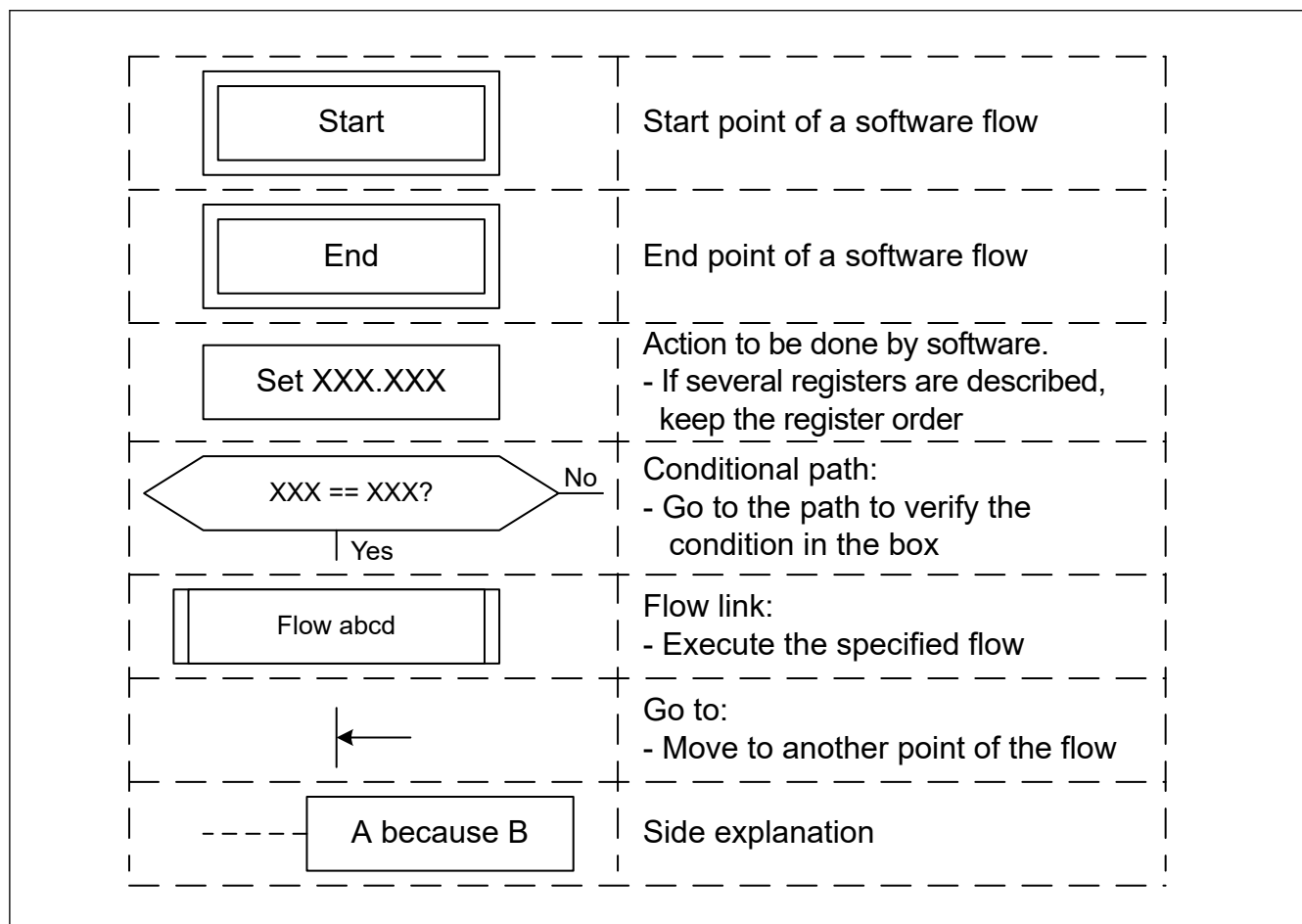
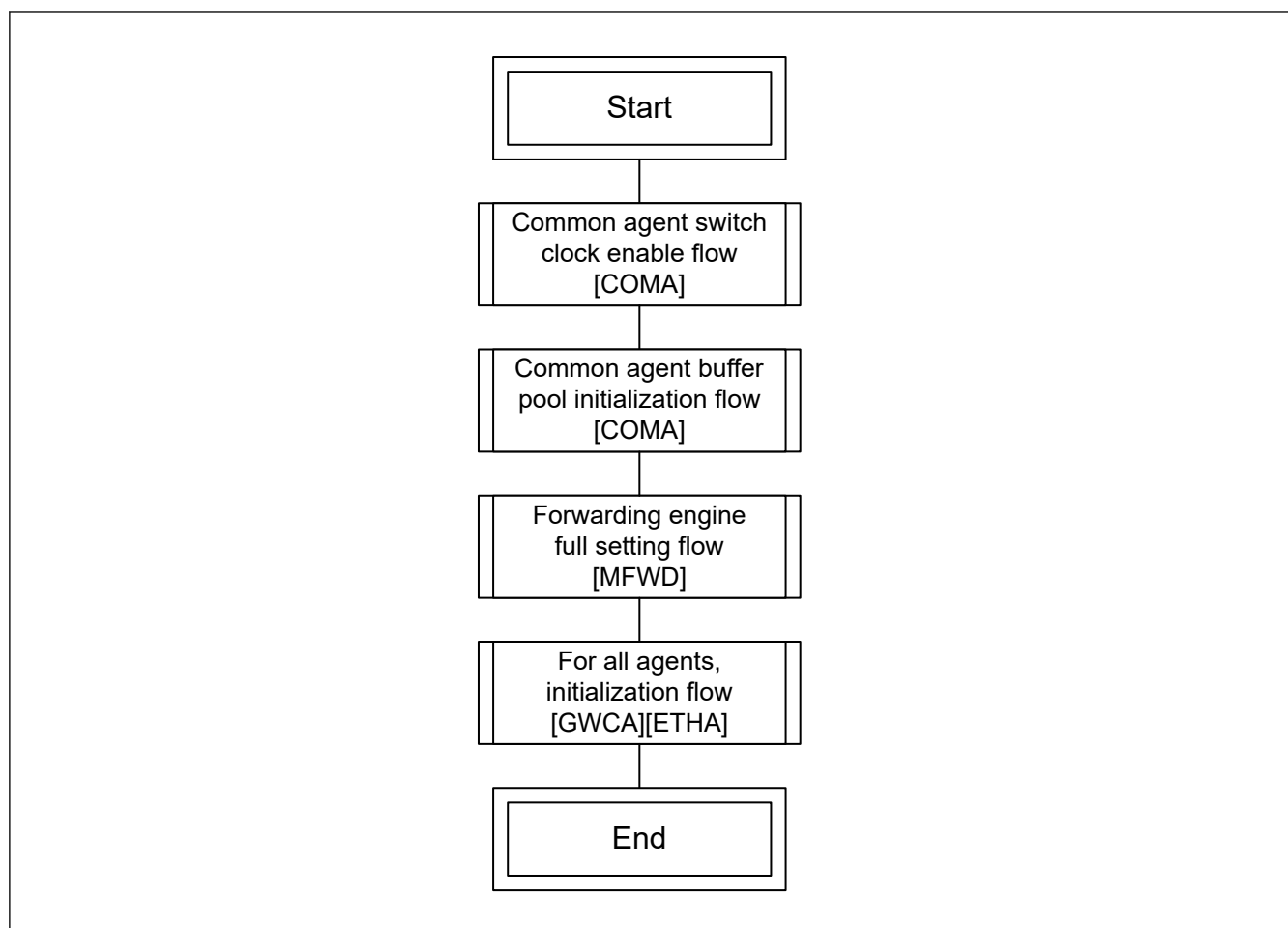


Figure 29.2 Software flow legend

### 29.3.1.2 Switch Initialization Flow

Figure 29.3 shows the switch initialization flow.

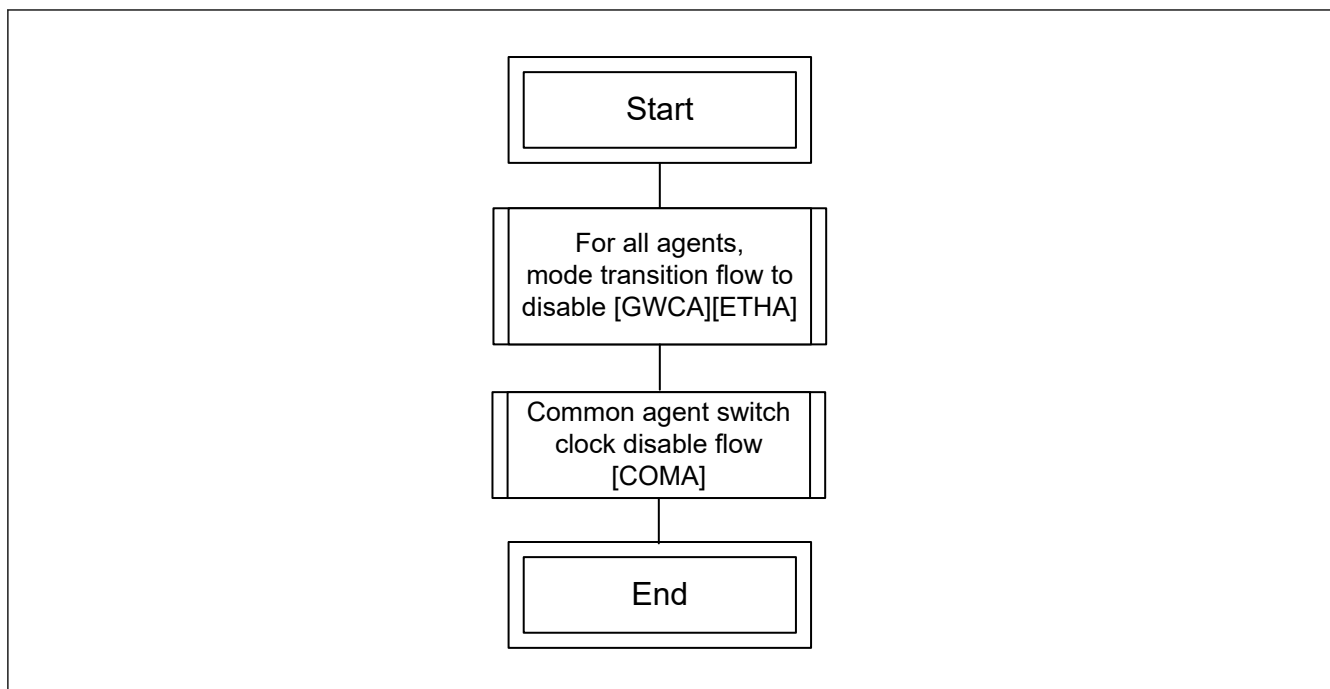




**Figure 29.3** Switch initialization flow

### 29.3.1.3 Switch Disable Flow

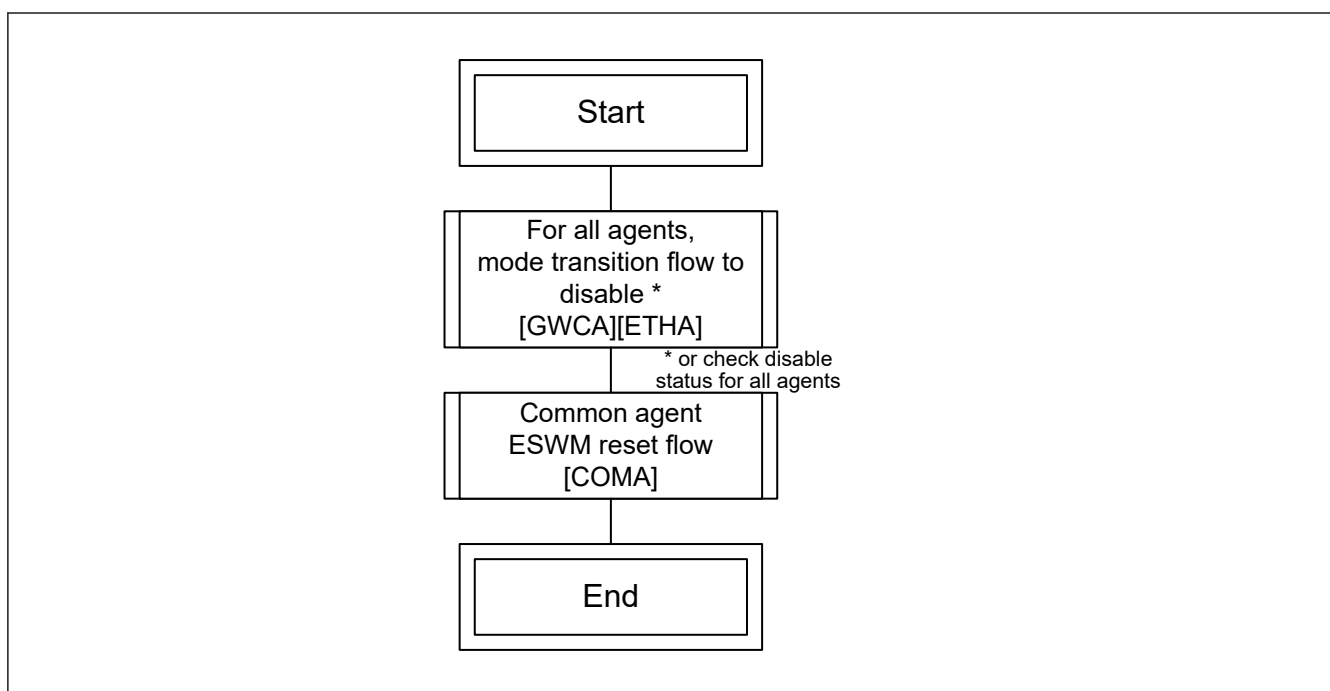
[Figure 29.4](#) shows the switch disable flow.



**Figure 29.4** Switch disable flow

### 29.3.1.4 Switch Reset Flow

Figure 29.5 shows the switch reset flow.

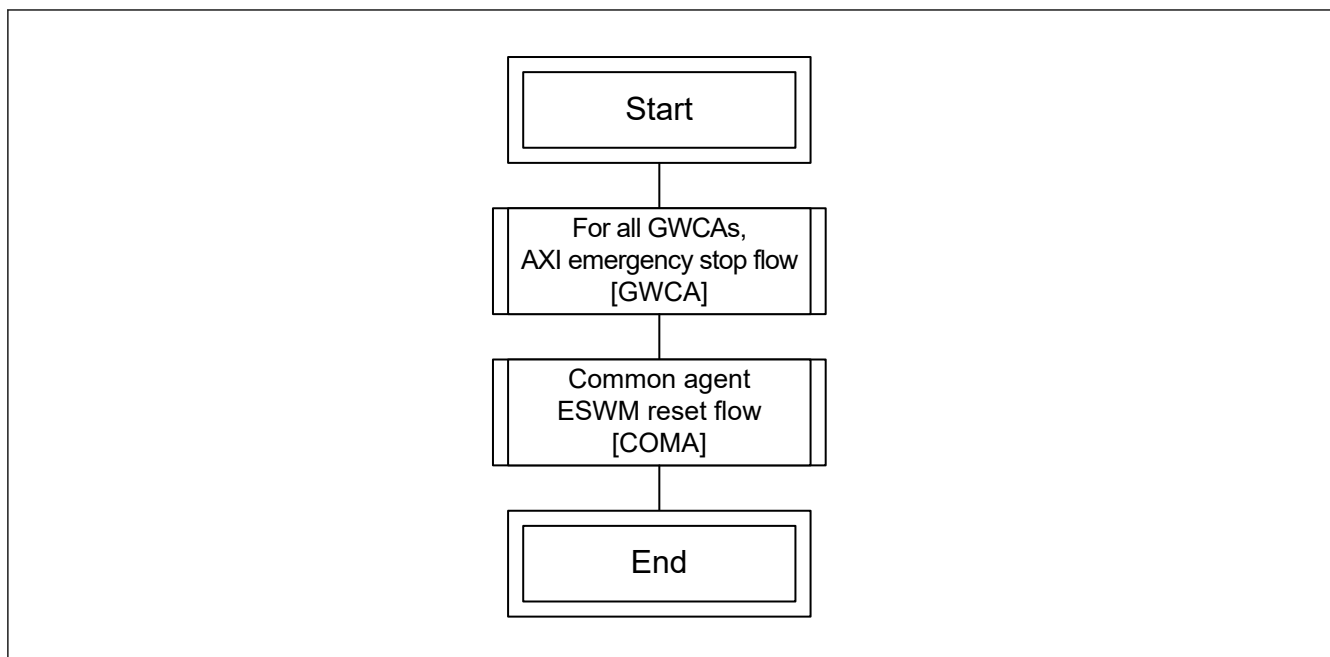


**Figure 29.5** Switch reset flow

### 29.3.1.5 Switch Emergency Reset Flow

Figure 29.6 shows the switch emergency reset flow used when [ETHA] is locked.

For example, in the case of MII, if an error occurs in the PHY and the MII PHY TX clock stops, the switch cannot send data and cannot release the pointer (locked on [ETHA] Operation mode). The state clears when the MII PHY TX clock is restarted, but use emergency reset if required.



**Figure 29.6** Switch emergency reset flow

### 29.3.1.6 Register Writable Without Software Flow

Table 29.3 describes the registers writable without software flow and in which mode they can be written.

**Table 29.3** Writable without software flow registers (✓ = writable, × = not writable)

| Register name | Writable anytime |
|---------------|------------------|
| TPEMIMC0      | ✓                |
| TPEMIMC1      | ✓                |
| TPEMIMC2      | ✓                |
| TPEMIMC3      | ✓                |
| TPEMIMC4      | ✓                |
| TPEMIMC5      | ✓                |
| TPEMIMC6t     | ✓                |
| TPEMIMC7t     | ✓                |
| TPICVC0       | ✓                |
| TPICVC1       | ✓                |
| TPICVC2       | ✓                |
| TPRCVC        | ✓                |
| TCVUC         | ✓                |

## 29.4 Functional Descriptions

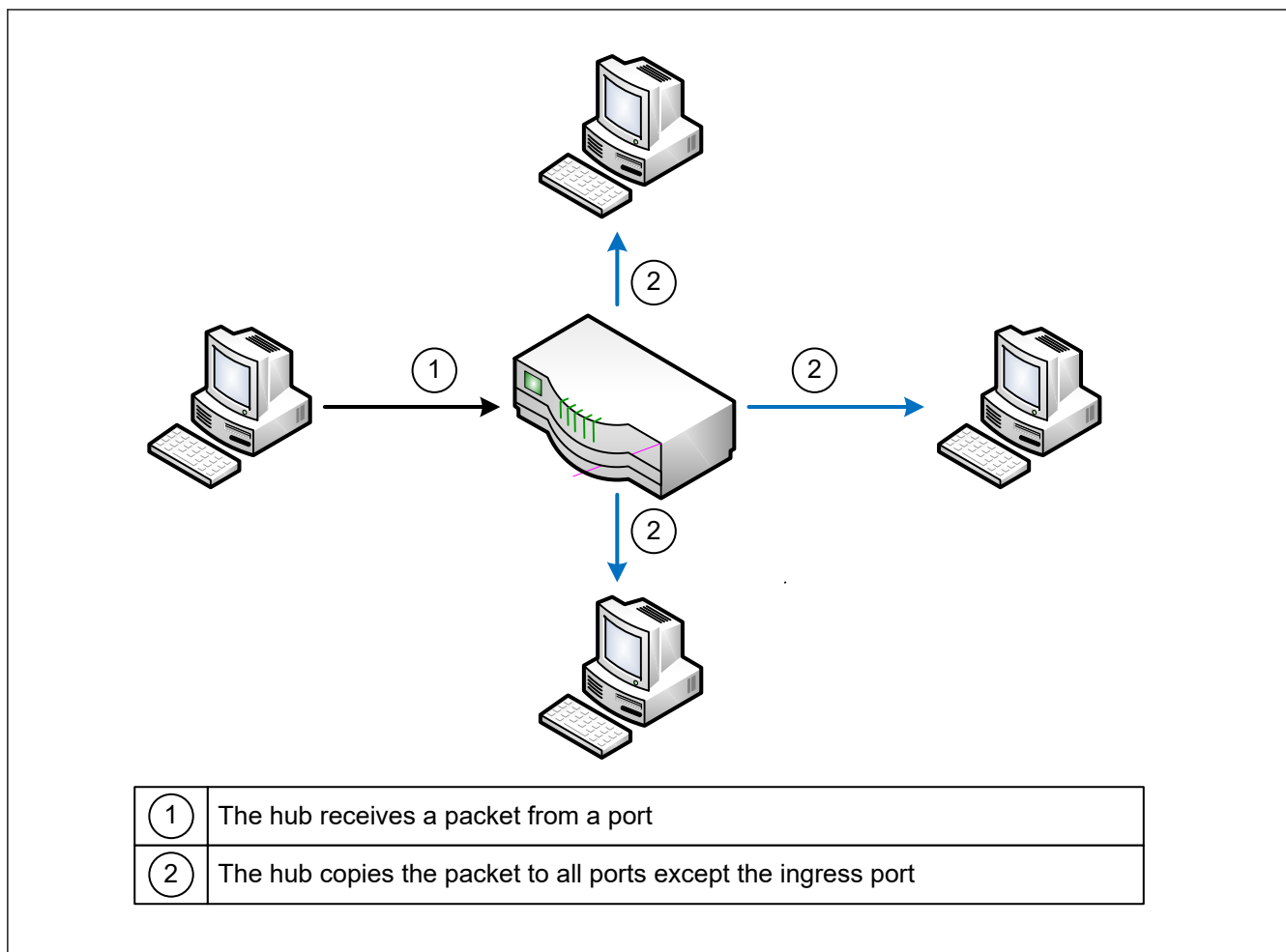
### 29.4.1 ESWM Forwarding Modes

#### 29.4.1.1 Hub

##### 29.4.1.1.1 Background

A hub is an ethernet network interconnection system (OSI) which only works on layer 1. It does not consider any kind of addressing and, consequently, is not aware of the kind of traffic that goes through it and can create security issues.

Figure 29.7 shows the hub operation.

**Figure 29.7 Hub operation**

Note: In Hub mode, the internal port [GWCA] behaves like an ethernet port. In the following configuration, the internal port receives all frames by AXI descriptor queue number 0 and transmits all frames by AXI descriptor queue number 1.

#### 29.4.1.1.2 Settings

Table 29.4 describes the setting required for ESWM in Hub mode. Specify this setting using the switch initialization flow described in [section 29.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

**Table 29.4 Hub settings (1 of 2)**

| Register name                           | Setting value                                    | Description                                                                                                                                 |
|-----------------------------------------|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Message Forwarding Engine [MFWD]</b> |                                                  |                                                                                                                                             |
| FWPBFCi.PBDV                            | All 1s except for FWPBFCi.PBDV which is set to 0 | All ports are set to send their incoming frames to all other ports                                                                          |
| <b>Ethernet Agent [ETHA]</b>            |                                                  |                                                                                                                                             |
| EAIRC                                   | 0                                                | All frames are directed to descriptor queue 0                                                                                               |
| EATDQDCq.DQD                            | If q = 0, set to 512<br>Else set to 0            | There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0. |
| EAVCC.VEM                               | 011b                                             | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                     |
| EATTFC.UT                               | 0                                                | Allows any TAG formats to go through even the unsupported ones                                                                              |

**Table 29.4 Hub settings (2 of 2)**

| Register name                   | Setting value                                                                                       | Description                                                                                                                                 |
|---------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Ethernet MAC [RMAC]</b>      |                                                                                                     |                                                                                                                                             |
| MPIC.PIS                        | User-defined                                                                                        | Set the PHY interface type                                                                                                                  |
| MPIC.LSC                        | User-defined                                                                                        | Set the PHY speed                                                                                                                           |
| MRAFC                           | 03C703C7H                                                                                           | Set MAC reception in promiscuous mode                                                                                                       |
| <b>Gateway CPU Agent [GWCA]</b> |                                                                                                     |                                                                                                                                             |
| GWIRC                           | 0                                                                                                   | All frames are directed to descriptor queue 0                                                                                               |
| GWRDQDCq.DQD                    | If q = 0, set to 512<br>Else set to 0                                                               | There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0. |
| GWVCC.VEM                       | 011                                                                                                 | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                     |
| GWTTC.UT                        | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                              |
| GWDCBAC0/1                      | User-defined                                                                                        | Set the LINKFIX table base address                                                                                                          |
| GWDCCi                          | For i = 1:<br>• Set DQT to 1<br>• Set OSID to user-defined.<br>Else:<br>• Set OSID to user-defined. | Queue 0 is used for data reception and queue 1 is used for data transmission. All other queues are not used.                                |

### 29.4.1.2 Layer 2 Switch

#### 29.4.1.2.1 Background

A layer 2 switch is an ethernet network interconnection system (OSI) which only works on layer 2 using MAC addresses. MAC addresses are used to filter and forward messages. A layer 2 switch has three main functions:

- Address learning: A switch can learn MAC addresses by looking at received frame source MAC addresses
- Address aging: A switch can suppress a MAC address if a frame from the corresponding source has not been received since a given period (usually 5 minutes)
- Frame forward/filter: A switch can forward or filter a frame based on its destination MAC address.

Figure 29.8 shows the layer 2 switch operation.

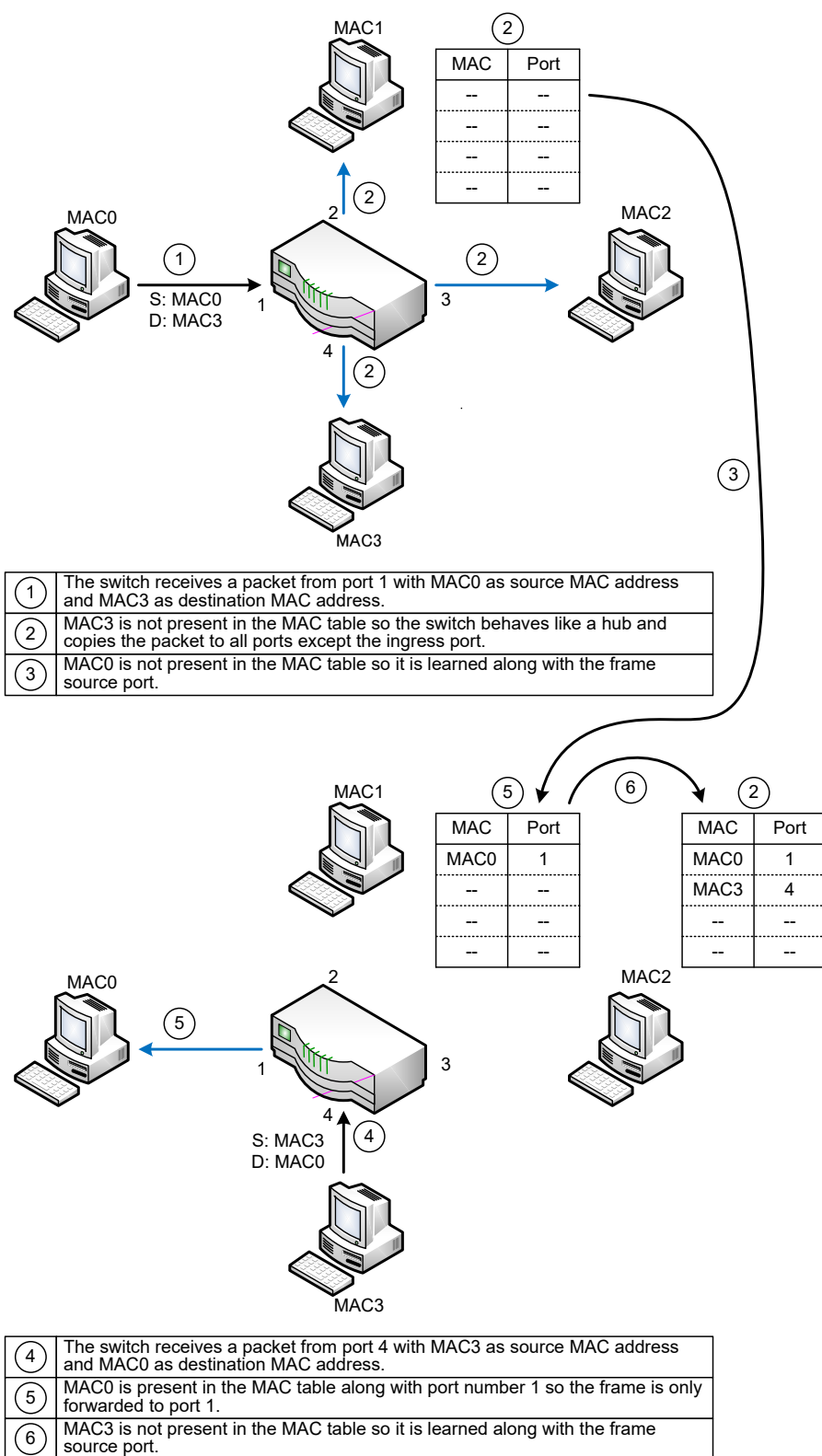


Figure 29.8 Layer 2 switch operation

### 29.4.1.2.2 Settings

Table 29.5 describes the setting required for ESWM in layer 2 switch mode. Specify this setting using the switch initialization flow described in [section 29.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

**Table 29.5 Layer 2 switch settings**

| Register name                           | Setting value                                                                                       | Explanation                                                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Message Forwarding Engine [MFWD]</b> |                                                                                                     |                                                                                                                                                                                                                   |
| FWPBFci.PBDV                            | All 1s except for FWPBFci.PBDV which is set to 0                                                    | All ports are set to send their incoming frames to all other ports                                                                                                                                                |
| FWPCi0.MACDSA                           | 1                                                                                                   | MAC destination search occurs for all ports                                                                                                                                                                       |
| FWPCi0.MACSSA                           | 1                                                                                                   | MAC source search occurs for all ports                                                                                                                                                                            |
| FWPCi0.MACHLA                           | 1                                                                                                   | MAC source hardware learning is enabled for all ports                                                                                                                                                             |
| FWPCi0.MACHMA                           | 1                                                                                                   | MAC source hardware migration is enabled for all ports                                                                                                                                                            |
| FWCLPRC.USMACLF                         | User-defined                                                                                        | Unknown source MAC addresses can be sent to CPU for learning monitoring                                                                                                                                           |
| FWCLPTC.LPCSD                           | 2                                                                                                   | Set the learning path queue number to 2                                                                                                                                                                           |
| FWCLPTC.LPCS                            | User-defined                                                                                        | Select which CPU to be used for learning monitoring                                                                                                                                                               |
| <b>Ethernet Agent [ETHA]</b>            |                                                                                                     |                                                                                                                                                                                                                   |
| EAIRC                                   | 0                                                                                                   | All frames are directed to descriptor queue 0                                                                                                                                                                     |
| EATDQDCq.DQD                            | If q = 0, set to 512<br>Else set to 0                                                               | There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.                                                                       |
| EAVCC.VEM                               | 011b                                                                                                | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| EATTFC.UT                               | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| <b>Ethernet MAC [RMAC]</b>              |                                                                                                     |                                                                                                                                                                                                                   |
| MPIC.PIS                                | User-defined                                                                                        | Set the PHY interface type                                                                                                                                                                                        |
| MPIC.LSC                                | User-defined                                                                                        | Set the PHY speed                                                                                                                                                                                                 |
| MRAFC                                   | 0x03C703C7                                                                                          | Set MAC reception in promiscuous mode                                                                                                                                                                             |
| <b>Gateway CPU Agent [GWCA]</b>         |                                                                                                     |                                                                                                                                                                                                                   |
| GWIRC                                   | 0                                                                                                   | All frames are directed to descriptor queue 0                                                                                                                                                                     |
| GWRDQDCq.DQD                            | If q = 0, set to 512<br>Else set to 0                                                               | There is no QoS, so all frames are stored in priority 0 queue. As a result, all the output descriptor queue memory is reserved for queue 0.                                                                       |
| GWVCC.VEM                               | 011b                                                                                                | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| GWTTC.UT                                | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| GWDCBAC0/1                              | User-defined                                                                                        | Set the LINKFIX table base address                                                                                                                                                                                |
| GWDCCi                                  | For i = 1:<br>• Set DQT to 1<br>• Set OSID to user-defined.<br>Else:<br>• Set OSID to user-defined. | Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used.<br>Note: For learning monitoring, use extended descriptors. |

Note: Any MAC address learned by hardware coming from an internal port [GWCA] is always learned along with AXI descriptor queue 0. As a result, always reserve AXI descriptor queue 0 for hardware learned MAC addresses that correspond to frames reception queue.

### 29.4.1.3 VLAN Aware Layer 2 Switch

#### 29.4.1.3.1 Background

A VLAN aware layer 2 switch is an ethernet network interconnection system (OSI) which works on layer 2 using MAC addresses and can reduce its broadcast domains in subdomains using VLAN IDs. A VLAN aware layer 2 switch has the same functions as a layer 2 switch but can also handle VLANs.

[Figure 29.9](#) shows the VLAN aware layer 2 switch operation.



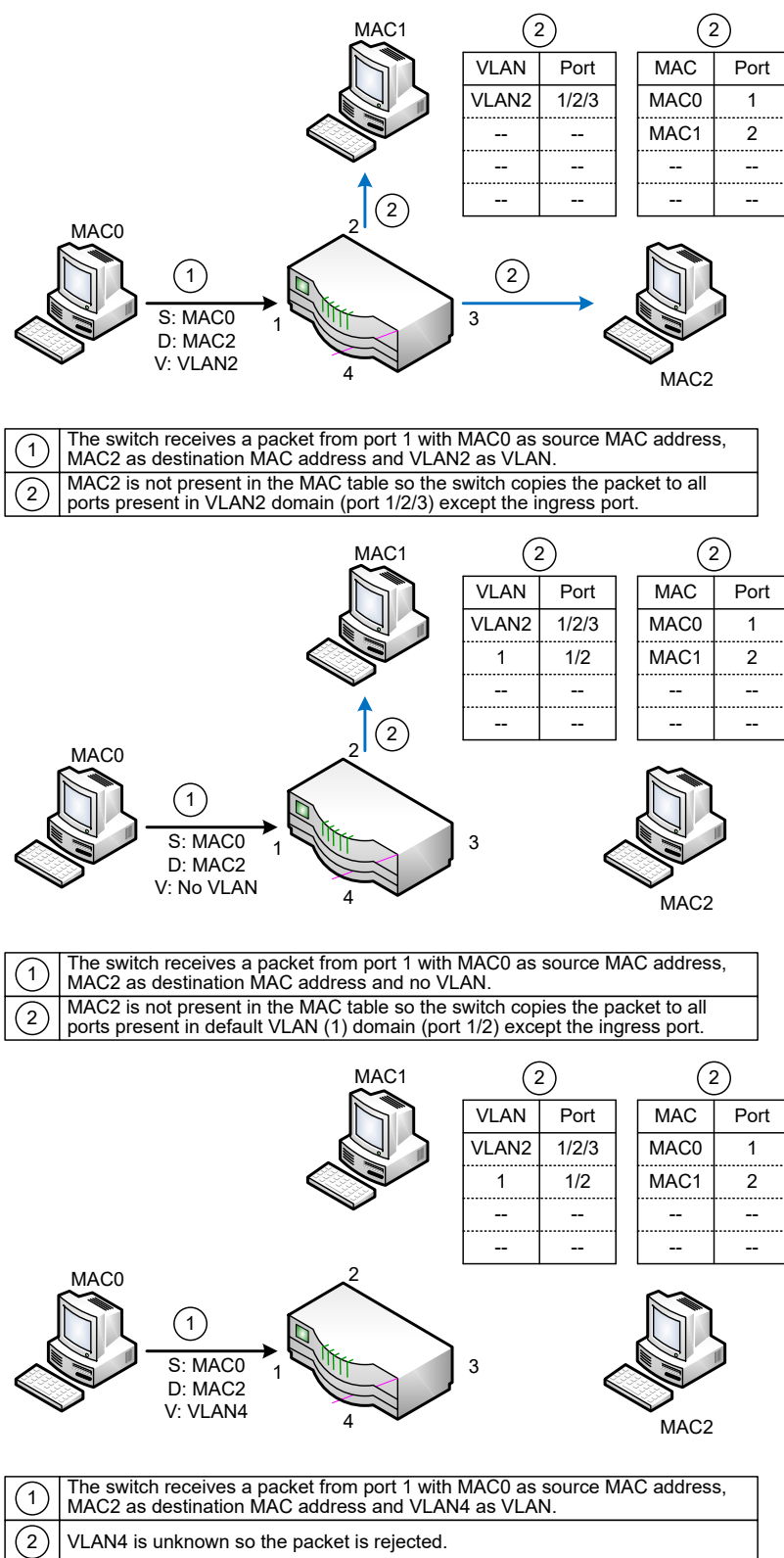


Figure 29.9 VLAN aware layer 2 switch operation

### 29.4.1.3.2 Settings

Table 29.6 describes the setting required for ESWM in VLAN aware layer 2 switch mode. Specify this setting using the switch initialization flow described in [section 29.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

**Table 29.6 VLAN aware layer 2 switch settings**

| Register name                           | Setting value                                                                                       | Explanation                                                                                                                                                                                                       |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Message forwarding engine [MFWD]</b> |                                                                                                     |                                                                                                                                                                                                                   |
| FWGC.SVM                                | 01                                                                                                  | Set ESWM in C-TAG mode                                                                                                                                                                                            |
| FWPBFCi.PBDV                            | All 1s except for FWPBFCi.PBDV which is set to 0                                                    | All ports are set to send their incoming frames to all other ports                                                                                                                                                |
| FWPCi0.MACDSA                           | 1                                                                                                   | MAC destination search occurs for all ports                                                                                                                                                                       |
| FWPCi0.MACSSA                           | 1                                                                                                   | MAC source search occurs for all ports                                                                                                                                                                            |
| FWPCi0.MACHLA                           | 1                                                                                                   | MAC source hardware learning is enabled for all ports                                                                                                                                                             |
| FWPCi0.MACHMA                           | 1                                                                                                   | MAC source hardware migration is enabled for all ports                                                                                                                                                            |
| FWPCi0.VLANSA                           | 1                                                                                                   | VLAN search occurs for all ports                                                                                                                                                                                  |
| FWPCi0.VLANRU                           | 1                                                                                                   | Unknown VLANs are rejected                                                                                                                                                                                        |
| FWCLPRC.USMACLF                         | User-defined                                                                                        | Unknown source MAC addresses can be sent to CPU for learning monitoring                                                                                                                                           |
| FWCLPRC.UVLANLF                         | User-defined                                                                                        | Unknown VLANs can be sent to CPU for learning                                                                                                                                                                     |
| FWCLPTC.LPCSD                           | 2                                                                                                   | Set the learning path queue number to 2                                                                                                                                                                           |
| FWCLPTC.LPCS                            | User-defined                                                                                        | Select which CPU to be used for learning monitoring                                                                                                                                                               |
| FWVLANTL0/1/2/3/4                       | User-defined                                                                                        | New VLANs can be added to the VLAN table.<br>For details, see <a href="#">Table 29.7</a> .                                                                                                                        |
| <b>Ethernet agent [ETHA]</b>            |                                                                                                     |                                                                                                                                                                                                                   |
| EATDQDCq.DQD                            | User-defined                                                                                        | The number of descriptors in descriptor queues can be set                                                                                                                                                         |
| EAVCC.VEM                               | 011                                                                                                 | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| EAVTC.CTV                               | 1                                                                                                   | Set the default VLAN value to 1                                                                                                                                                                                   |
| EATTFC.UT                               | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| <b>RMAC [RMAC]</b>                      |                                                                                                     |                                                                                                                                                                                                                   |
| MPIC.PIS                                | User-defined                                                                                        | Set the PHY interface type                                                                                                                                                                                        |
| MPIC.LSC                                | User-defined                                                                                        | Set the PHY speed                                                                                                                                                                                                 |
| MRAFC                                   | 03C703C7H                                                                                           | Set MAC reception in promiscuous mode                                                                                                                                                                             |
| <b>GWCA [GWCA]</b>                      |                                                                                                     |                                                                                                                                                                                                                   |
| GWRDQDCq.DQD                            | User-defined                                                                                        | The number of descriptors in descriptor queues can be set                                                                                                                                                         |
| GWVCC.VEM                               | 011                                                                                                 | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| GWVTC.CTV                               | 1                                                                                                   | Set the default VLAN value to 1                                                                                                                                                                                   |
| GWTTFC.UT                               | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| GWDCBAC0/1                              | User-defined                                                                                        | Set the LINKFIX table base address                                                                                                                                                                                |
| GWDCCi                                  | For i = 1:<br>• Set DQT to 1<br>• Set OSID to user-defined.<br>Else:<br>• Set OSID to user-defined. | Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used.<br>Note: For learning monitoring, use extended descriptors. |

**Table 29.7 VLAN learning in VLAN table**

| Field name VLAN | Setting value                       | Explanation                                                                   |
|-----------------|-------------------------------------|-------------------------------------------------------------------------------|
| SL              | 0                                   | No secure entry used                                                          |
| HLD             | 0                                   | Hardware learning for MAC addresses not disabled for any VLAN                 |
| SLV             | Set to 1 for ports included in VLAN | Only authorize to receive corresponding VLAN from ports included in this VLAN |
| DV              | Set to 1 for ports included in VLAN | Only authorize to send corresponding VLAN to ports included in this VLAN      |
| CSD             | 0                                   | Frames received by CPUs on sub-destination 0                                  |
| CME             | 0                                   | No mirroring                                                                  |
| EME             | 0                                   | No mirroring                                                                  |
| IPU             | 0                                   | No frame priority update                                                      |
| IPV             | 0                                   | No frame priority update                                                      |

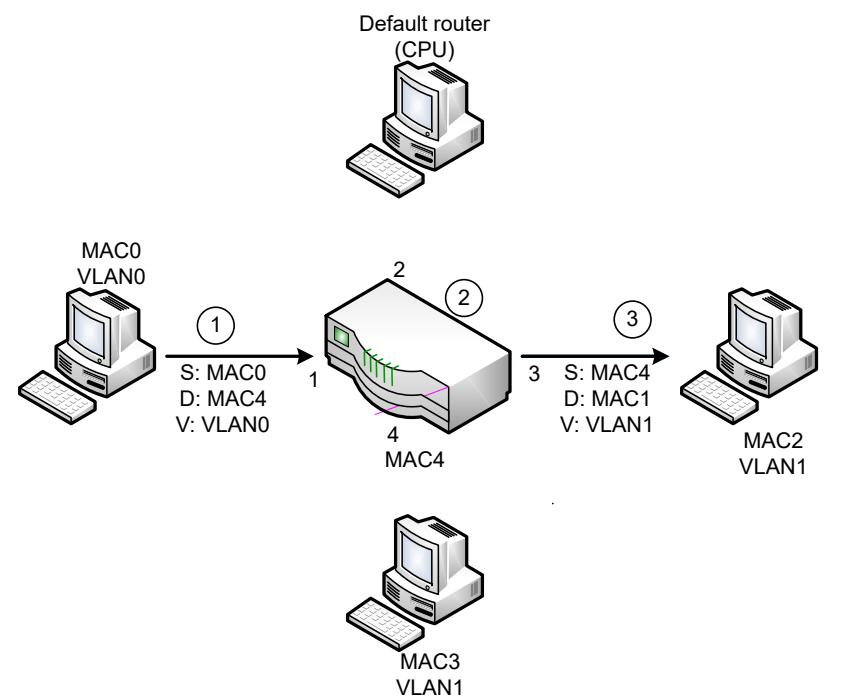
Note: When both VLAN and destination MAC addresses are found in forwarding tables and corresponding frame is forwarded to an internal port, the AXI descriptor queue (CPU sub-destination) chosen for forwarding is the one set in the MAC table. In this example, because both are set to 0, it raises no issue but, when AXI descriptor queues are used for virtualization, it should be taken in account during system design.

## 29.4.1.4 Layer 3 Switch

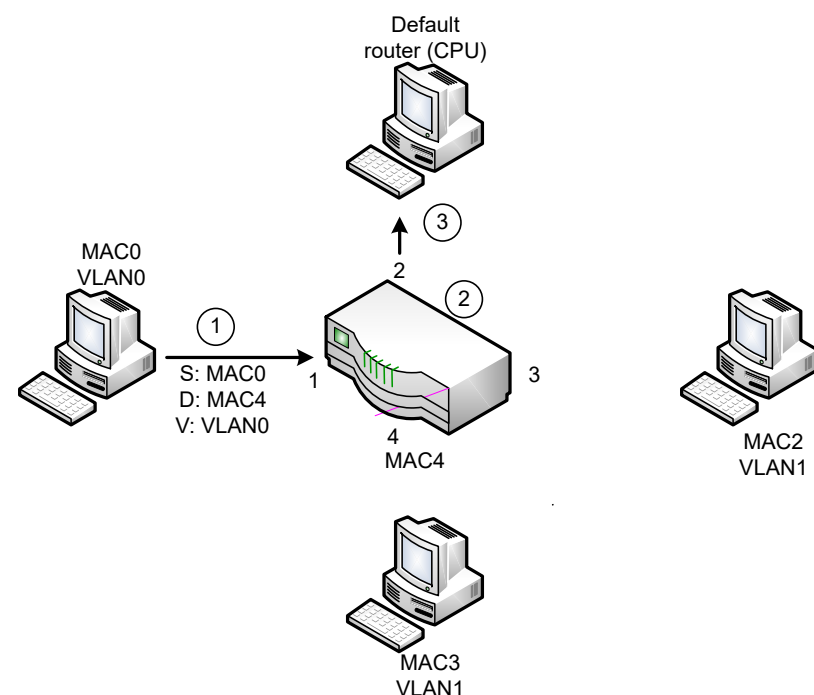
### 29.4.1.4.1 Background

A layer 3 switch is an ethernet network interconnection system (OSI) which works on layer 3 using IP addresses. Because ESWM can only perform perfect matches by stream forwarding, when developing a layer 3 switch, it is recommended to use perfect filters [FWD].

[Figure 29.10](#) shows the layer 3 switch operation.



|   |                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------|
| ① | The switch receives a packet from port 1 with MAC0 as source MAC address, MAC4 as destination MAC address and VLAN0 as VLAN. |
| ② | The packet IP address matches the cascade filter and should be routed to port 3.                                             |
| ③ | The switch updates MAC destination to MAC1, MAC source to MAC4 and VLAN to VLAN1 and sends the frame to port 3.              |



|   |                                                                                                                              |
|---|------------------------------------------------------------------------------------------------------------------------------|
| ① | The switch receives a packet from port 1 with MAC0 as source MAC address, MAC4 as destination MAC address and VLAN0 as VLAN. |
| ② | The packet IP address is unknown so routing cannot happen.                                                                   |
| ③ | The switch forwards the frame to the default router (CPU) without update for routing.                                        |

Figure 29.10 Layer 3 switch operation

### 29.4.1.4.2 Settings

Table 29.8 describes the setting required for ESWM in layer 3 switch mode. Specify this setting using the switch initialization flow described in [section 29.3.1.2. Switch Initialization Flow](#). Any register not described in the table should be kept to its initial value, so it does not need to be set. After using the switch initialization flow, any used port should be moved to Operation mode when required.

**Table 29.8 Layer 3 switch settings**

| Register name                   | Setting value                                                                                       | Explanation                                                                                                                                                                                                       |
|---------------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Forwarding engine [MFWD]</b> |                                                                                                     |                                                                                                                                                                                                                   |
| FWGC.SVM                        | 01                                                                                                  | Set ESWM in C-TAG mode                                                                                                                                                                                            |
| FWPBFCi.PBDV                    | All 1s except for FWPBFCi.PBDV which is set to 0                                                    | All ports are set to send their incoming frames to all other ports                                                                                                                                                |
| FWPCi0.MACDSA                   | 1                                                                                                   | MAC destination search occurs for all ports                                                                                                                                                                       |
| FWPCi0.MACSSA                   | 1                                                                                                   | MAC source search occurs for all ports                                                                                                                                                                            |
| FWPCi0.MACHLA                   | 1                                                                                                   | MAC source hardware learning is enabled for all ports                                                                                                                                                             |
| FWPCi0.MACHMA                   | 1                                                                                                   | MAC source hardware migration is enabled for all ports                                                                                                                                                            |
| FWPCi0.VLANSA                   | 1                                                                                                   | VLAN search occurs for all ports                                                                                                                                                                                  |
| FWCLPRC.USMACLF                 | User-defined                                                                                        | Unknown source MAC addresses can be sent to CPU for learning monitoring                                                                                                                                           |
| FWCLPRC.UVLANLF                 | 1                                                                                                   | Unknown VLANs are sent to CPU for learning                                                                                                                                                                        |
| FWCLPTC.LPCSD                   | 2                                                                                                   | Set the learning path queue number to 2                                                                                                                                                                           |
| FWCLPTC.LPCS                    | User-defined                                                                                        | Select which CPU to be used for learning monitoring                                                                                                                                                               |
| FWVLANTL0/1/2/30/4              | User-defined                                                                                        | New VLANs can be added to the VLAN table.<br>For details, see <a href="#">Table 29.7</a> .                                                                                                                        |
| <b>Ethernet agent [ETHA]</b>    |                                                                                                     |                                                                                                                                                                                                                   |
| EATDQDCq.DQD                    | User-defined                                                                                        | The number of descriptors in descriptor queues can be set                                                                                                                                                         |
| EAVCC.VEM                       | 011                                                                                                 | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| EAVTC.CTV                       | 1                                                                                                   | Set the default VLAN value to 1                                                                                                                                                                                   |
| EATTFC.UT                       | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| <b>RMAC [RMAC]</b>              |                                                                                                     |                                                                                                                                                                                                                   |
| MPIC.PIS                        | User-defined                                                                                        | Set the PHY interface type                                                                                                                                                                                        |
| MPIC.LSC                        | User-defined                                                                                        | Set the PHY speed                                                                                                                                                                                                 |
| MRAFC                           | 03C703C7H                                                                                           | Set MAC reception in promiscuous mode                                                                                                                                                                             |
| <b>GWCA [GWCA]</b>              |                                                                                                     |                                                                                                                                                                                                                   |
| GWRDQDCq.DQD                    | User-defined                                                                                        | The number of descriptors in descriptor queues can be set                                                                                                                                                         |
| GWVCC.VEM                       | 011                                                                                                 | Allows SC-tagged frames and C-tagged frames to go through transparently                                                                                                                                           |
| GWVTC.CTV                       | 1                                                                                                   | Set the default VLAN value to 1                                                                                                                                                                                   |
| GWTTC.UT                        | 0                                                                                                   | Allows any TAG formats to go through even the unsupported ones                                                                                                                                                    |
| GWDCBAC0/1                      | User-defined                                                                                        | Set the LINKFIX table base address                                                                                                                                                                                |
| GWDCCi                          | For i = 1:<br>• Set DQT to 1<br>• Set OSID to user-defined.<br>Else:<br>• Set OSID to user-defined. | Queue 0 is used for data reception, queue 1 is used for data transmission and queue 2 is used for learning monitoring. All other queues are not used.<br>Note: For learning monitoring, use extended descriptors. |

**Table 29.9 VLAN learning in VLAN table**

| Field name VLAN | Setting value                       | Explanation                                                                   |
|-----------------|-------------------------------------|-------------------------------------------------------------------------------|
| SL              | 0                                   | No secure entry used                                                          |
| HLD             | 0                                   | Hardware learning for MAC addresses not disabled for any VLAN                 |
| SLV             | Set to 1 for ports included in VLAN | Only authorize to receive corresponding VLAN from ports included in this VLAN |
| DV              | Set to 1 for ports included in VLAN | Only authorize to send corresponding VLAN to ports included in this VLAN      |
| CSD             | 0                                   | Frames received by CPUs on sub-destination 0                                  |
| CME             | 0                                   | No mirroring                                                                  |
| EME             | 0                                   | No mirroring                                                                  |
| IPU             | 0                                   | No frame priority update                                                      |
| IPV             | 0                                   | No frame priority update                                                      |

### 29.4.1.5 AVB scheduling

#### 29.4.1.5.1 Forwarding and Queuing Enhancements for Time-Sensitive Streams [802.1Qav]

### 29.4.2 Precautions

- ESWM can support max  $(([\text{Unused pointer number}] \times 128) / 4)$  bytes frame (supported maximum frame size). A buffer overflow occurs when the transmit frame size exceeds the supported maximum size.  $[\text{Unused pointer number}] = 512 - (\text{Activated port number} \times 6)$ .
  - Set to [RMAC] MRFSC.EMXS smaller than maximum frame size for Ethernet ports
  - Do not transmit larger than maximum frame size for CPU ports.
- GWCA can only transmit/receive frames with maximum size of 63488 bytes for the first frame. For the upcoming frames, the maximum frame size can be 65535 bytes provided LINK/LINKFIX descriptors were not used in between the queue.
- When using Cut through, use it so as not to cause a buffer overflow. The following usage is recommended:
  - Only use from-one-to-one communication.  
(Example of prohibition : Multicast forwarding, to Ethernet mirroring, Loopback forwarding).
  - Prohibition of the function to stop transmission. Use flow control on reception port.  
(Example of prohibition : TX queue pause, CBS, TAS).
- The overall specifications of ESWM are determined on the frequency of the operating clock.  
See the following table for details to select the operating clock.

| Selection        |         |                    | Definition                                                   |                          |
|------------------|---------|--------------------|--------------------------------------------------------------|--------------------------|
| Local RAM size*1 | Clk*1   | Internal bandwidth | Ethernet & CPU total bandwidth                               | Throughput (Ether2Ether) |
| 64 KB            | 250 MHz | 32 Gbps            | MAX 1G bps (Exp 100M Ether 2 port + 100M CPU 1 port) × TX/RX | 800 to 2000 ns           |
| 64 KB            | 50 MHz  | 6.4 Gbps           | MAX 1G bps (Exp 100M Ether 2 port + 100M CPU 1 port) × TX/RX | 2000 to 6000 ns          |
| 64 KB            | 250 MHz | 32 Gbps            | MAX 3G bps (Exp 1G Ether 2 port + 1G CPU 1 port) × TX/RX     | 400 to 600 ns            |
| 64 KB            | 150 MHz | 19.2 Gbps          | MAX 3G bps (Exp 1G Ether 2 port + 1G CPU 1 port) × TX/RX     | 500 to 800 ns            |

Note: At the frame end (forwarding source port) - frame start (forwarding destination port). When measuring, there is no dependency on the payload length of the frame.

Note: The frame start (forwarding destination port) is up to the point where the preamble starts to output.

Note 1. Specifications less than these are prohibited.  
Throughput measurement conditions:

- From frame end (forwarding source port) to frame start (forwarding destination port)
- The forwarding (destination port) is not busy
- L2 forwarding unicast (1:1 forwarding).

## 29.5 Operation

### 29.5.1 Internal Clocks

Table 29.10 lists ESWM internal clocks. These clocks are supplied by the Clock Generation Circuit (CGC). Set the clock before starting the ESWM operation.

**Table 29.10 Configurable range of clocks**

| VDD Voltage [V]         | Speed [Mbps] | ICLK [MHz] | PCLKA [MHz] | ESWCLK [MHz]            | ESWPHYCLK [MHz] |
|-------------------------|--------------|------------|-------------|-------------------------|-----------------|
| VSCR_1, voltage range 1 | 1000         | 150 to 250 | up to 125   | 150 to 250              | 500             |
|                         | 100/10       | 50 to 250  | up to 125   | 50 to 250* <sup>1</sup> | —               |
| VSCR_2, voltage range 2 | 1000         | 100 to 150 | up to 75    | 150 to 225              | 500             |
|                         | 100/10       | 50 to 150  | up to 75    | 50 to 225* <sup>1</sup> | —               |

Note 1. When using products in the HLQFP176 package, set ESWCLK to 125 MHz or less.

Clock restrictions are as follows:

- $ICLK \times 1.5 \geq ESWCLK$
- $ESWCLK > PCLKA$
- $ICLK > PCLKA$ .

### 29.5.2 PHY Interface

#### 29.5.2.1 PHY Interface Select

Table 29.11 lists the configuration settings for the ESWM PHY interface. Set the MIICRn.MIISEL[1:0] bits first before setting the MPIC register in RMACn.

**Table 29.11 Configuration settings for ESWM PHY interface**

| Type  | Speed    | RMACn.MPIC |             | MIICRn      |
|-------|----------|------------|-------------|-------------|
|       |          | LSC[2:0]   | PIS[2:0]    | MIISEL[1:0] |
| MII   | 10 Mbps  | 000b       | 000b (MII)  | 00b         |
| MII   | 100 Mbps | 001b       | 000b (MII)  | 00b         |
| GMII  | 1 Gbps   | 010b       | 010b (GMII) | 00b         |
| RGMII | 10 Mbps  | 000b       | 000b (MII)  | 01b         |
| RGMII | 100 Mbps | 001b       | 000b (MII)  | 01b         |
| RGMII | 1 Gbps   | 010b       | 010b (GMII) | 01b         |
| RMII  | 10 Mbps  | 000b       | 000b (MII)  | 10b         |
| RMII  | 100 Mbps | 001b       | 000b (MII)  | 10b         |

#### 29.5.2.2 MII/GMII Interface

When MIICRn.MIISEL[1:0] = 00b, MII/GMII interface is selected. GTX\_CLK is generated by dividing ESWPHYCLK by 4.

#### 29.5.2.3 RGMII Interface

ESWM can be converted from an external RGMII to an internal GMII/MII. The RGMII interface is enabled when MIICRn.MIISEL[1:0] = 01b. Since the initial state of the ESWM PHY interface is MII/GMII, it becomes operational after

the reset is released with the MIIRR.RGRST bit. At this time, TXC is generated by dividing ESUPHYCLK. MIICRn. Set MIISEL[1:0] to 01b before a release reset.

### 29.5.2.3.1 RGMII Internal Delay Mode

The RGMII interface supports internal delay mode for RGMII<sub>n</sub>\_TXC.

### 29.5.2.4 RMII Interface

ESWM can be converted from an external RMII to an internal MII. The RMII interface is enabled when MIICRn.MIISEL[1:0] = 10b. Since the initial state of the ESWM PHY interface is MII/GMII, it becomes operational after the reset is released with the MIIRR.RMRST bit. Set MIICRn.MIISEL[1:0] to 10b before a release reset.

## 29.5.3 ESWM Core Function

See the following chapters:

- [section 30, Ethernet Message Forwarding Engine \(MFWD\)](#)
- [section 30.5.4. Fabric \(MFAB\)](#)
- [section 31, Ethernet Common Agent \(COMA\)](#)
- [section 32, Ethernet Agent \(ETHA\)](#)
- [section 33, Ethernet MAC \(RMAC\)](#)
- [section 34, Ethernet CPU Agent \(GWCA\).](#)

## 29.5.4 gPTP and PTP Pulse Generator

See the following chapters:

- [section 35, Ethernet Generic PTP Timer \(GPTP\)](#)
- [section 35.6. Pulse Output Timer Module \(plsotim\).](#)

## 29.5.5 TAS Gate Status Monitor

The signal to be output to the ET\_TAS\_STAn pin can be selected using the TASSTMS register. [Table 29.12](#) lists the TASSTMS register select table.

**Table 29.12**

| MSSn[4:0]        | Module | Signal                                                   |
|------------------|--------|----------------------------------------------------------|
| 00000b to 01000b | ETHA0  | eha_race_tas_gate_state[0] to eha_race_tas_gate_state[8] |
| 10000b to 11000b | ETHA1  | eha_race_tas_gate_state[0] to eha_race_tas_gate_state[8] |

For details of eha\_race\_tas\_gate\_state[8:0], see [section 32, Ethernet Agent \(ETHA\)](#).

## 29.6 Interrupt and Event

**Table 29.13** Interrupt and event output (1 of 2)

| Symbol      | Source flag                                                                            | Description                | Type  |
|-------------|----------------------------------------------------------------------------------------|----------------------------|-------|
| ETHER_FWEI  | MFWD.FWEIS                                                                             | Forwarding Error Interrupt | Level |
| ETHER_CAEI  | COMA.CAEIS                                                                             | Common Error Interrupt     | Level |
| ETHER_GWEI  | GWCA0.GWEIS0 (other than GWEIS0.TDFES),<br>GWCA0.GWEIS1, GWCA0.GWEIS4,<br>GWCA0.GWEIS5 | GWCA0 Error Interrupt      | Level |
| ETHER_EAEI0 | ETHA0.EAEIS0, ETHA0.EAEIS1, ETHA0.EAEIS2,<br>RMAC0.MEIS                                | ETHA0 Error Interrupt      | Level |
| ETHER_EAEI1 | ETHA1.EAEIS0, ETHA1.EAEIS1, ETHA1.EAEIS2,<br>RMAC1.MEIS                                | ETHA1 Error Interrupt      | Level |



**Table 29.13** Interrupt and event output (2 of 2)

| Symbol       | Source flag                                                              | Description                  | Type  |
|--------------|--------------------------------------------------------------------------|------------------------------|-------|
| ETHER_PTPI0  | GTP.PTPI0                                                                | gPTP Status Interrupt 0      | Level |
| ETHER_PTPI1  | GTP.PTPI1                                                                | gPTP Status Interrupt 1      | Level |
| ETHER_FWSI   | MFWD.FWMIS                                                               | Forwarding Status Interrupt  | Level |
| ETHER_CAMI   | COMA.CAMIS                                                               | Common Status Interrupt      | Level |
| ETHER_EASI0  | RMAC0.MMIS                                                               | ETHA0 Status Interrupt       | Level |
| ETHER_EASI1  | RMAC1.MMIS                                                               | ETHA1 Status Interrupt       | Level |
| ETHER_GWDI7  | GWCA0.GWDIDS, GWCA0.GWEIS2,<br>GWCA0.GWEIS3 with mapped by TOP.TPEMIMC7t | GWCA Data Interrupt          | Level |
| ETHER_GWDI6  |                                                                          |                              |       |
| ETHER_GWDI5  |                                                                          |                              |       |
| ETHER_GWDI4  |                                                                          |                              |       |
| ETHER_GWDI3  |                                                                          |                              |       |
| ETHER_GWDI2  |                                                                          |                              |       |
| ETHER_GWDI1  |                                                                          |                              |       |
| ETHER_GWDI0  |                                                                          |                              |       |
| ETHER_TSDI1  | GWCA0.GWTSDIS, GWCA0.GWEIS0.TDFES with<br>mapped by TOP.TPMIMC6t         | Time Stamp Data Interrupt    | Level |
| ETHER_TSDI0  |                                                                          | Time Stamp Data Interrupt    | Level |
| ETHER_MDIO0  | RMAC0.MMIS1                                                              | MDIO Interrupt 0             | Level |
| ETHER_MDIO1  | RMAC1.MMIS1                                                              | MDIO Interrupt 1             | Level |
| ETHER_RMPIO  | RMAC0.MMIS.LPIAIS, RMAC0.MMIS.LPIDIS,<br>RMAC0.MMIS.MPDIS                | RMAC0 PHY Interrupt          | Level |
| ETHER_RMPIO1 | RMAC1.MMIS.LPIAIS, RMAC1.MMIS.LPIDIS,<br>RMAC1.MMIS.MPDIS                | RMAC1 PHY Interrupt          | Level |
| GPTP_MATCH0  | —                                                                        | Media Clock Recovery Match 0 | Pulse |
| GPTP_MATCH1  | —                                                                        | Media Clock Recovery Match 1 | Pulse |
| GPTP_PTPOUT0 | —                                                                        | PTP Pulse Out 0              | Pulse |
| GPTP_PTPOUT1 | —                                                                        | PTP Pulse Out 1              | Pulse |
| GPTP_PTPOUT2 | —                                                                        | PTP Pulse Out 2              | Pulse |
| GPTP_PTPOUT3 | —                                                                        | PTP Pulse Out 3              | Pulse |

**Table 29.14** Event input

| Symbol    | Event source | Description                      |
|-----------|--------------|----------------------------------|
| ELC_GPTP0 | ELSR31       | gPTP Media clock capture event 0 |
| ELC_GPTP1 | ELSR32       | gPTP Media clock capture event 1 |

The gPTP media clock capture can select event source using the PTP Capture Event Select register.

## 29.7 Usage Notes

### 29.7.1 Settings for the Module-Stop Function

The MSTPC30 bit in Module Stop Control Register C (MSTPCRC) can enable or disable the ESWM module operation. The modules are initially stopped after reset. Releasing the module-stop state enables access to the registers. For details, see [section 11, Low Power Mode](#).

### 29.7.2 Transition to Software Standby Mode

ESWM should run Switch disable flow to disable it before transitioning to software standby.

If RGMII interface or RMII interface is selected, these interfaces need to be reset after Switch disable flow by the MIIRR register.

### 29.7.3 Return from Software Standby Mode

After returning from Software Standby mode, ESWM should run the switch initialization flow to initialize it.

### 29.7.4 Transition to ESWM Power Domain off by Power Gating

ESWM should run Switch disable flow to disable it before transitioning to the ESWM power domain off.

If RGMII interface or RMII interface is selected, these interfaces need to be reset after switch disable flow by the MIIRR register.

### 29.7.5 Transition to ESWM Power Domain on by Power Gating

After the ESWM power domain is on, ESWM should run the switch initialization flow to initialize it.